EMV® Terminal Type Approval Bulletin No. 268
Second Edition, December 2023

Terminal Level 2 Test Cases
4.4a Update

Applicability

This Bulletin applies to:

- *EMV® Terminal Type Approval - Level 2 Test Cases – version 4.4a*

Related Documents

This Bulletin should be read in conjunction with:

- *EMV® Terminal Type Approval - Level 2 Test Cases – version 4.4a*

Effective Date

- January 1st, 2024

Description

Based on EMVCo's continued review of *EMV® Terminal Type Approval Level 2 Test Cases v4.4a,* the following updates are necessary, and shall be applied to any type approval session beginning on, or after, the effective date cited above.

Update to EMV® Terminal Type Approval - Level 2 Test Cases – version 4.4a

Section 5 – Test Settings Updates

Section 5.5 – TVR & TSI un-setting bits checking

<u>TVR ICS-bitmask</u>

The ICS-bitmask value is defined by the tool at the start of a kernel configuration test session. This bitmask is depending on the supported options of this kernel configuration (listed in the ICS) and stay at the same value during the whole kernel configuration testing. It must be recalculated based on the supported options for each kernel configuration under test.
The ICS-bitmask is limited to some TVR bits and are depending on the related ICS question(s) other TVR bit shall have the value ‘0’ in the bitmask:

TVR	TVR Name	Related ICS question of the configuration	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed	SDA and DDA = NO	1

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B1b6	ICC data missing		0
B1b5	Card appears on terminal exception file	Exception File = NO	1
B1b4	DDA failed	SDA and DDA = NO	1
B1b3	CDA failed	CDA = NO	1
B1b2	SDA Selected	SDA and DDA = NO	1
B1b1	XDA Selected	XDA = No	1
B2b8	ICC and terminal have different application versions		0
B2b7	Expired application		0
B2b6	Application not yet effective		0
B2b5	Requested service not allowed for card product		0
B2b4	New card	Velocity Checking = NO	1
B2b3	RFU	Always set	1
B2b2	RFU or Biometric not performed or not successful	Biometric = No	1
B2b1	RFU or Biometric template format supported	Biometric = No	1
B3b8	Cardholder verification was not successful		0
B3b7	Unrecognised CVM		0
B3b6	PIN Try Limit exceeded	Offline Enciphered PIN AND Plaintext PIN = NO	1
B3b5	PIN entry required and PIN pad not present or not working		0
B3b4	PIN entry required, PIN pad present, but PIN was not entered	Offline Enciphered PIN AND Plaintext PIN AND Online Enciphered PIN = NO	1
B3b3	Online PIN entered	Online Enciphered PIN = NO	1
B3b2	RFU or Biometric required but Biometric capture device not working	Biometric = No	1
B3b1	RFU or Biometric required, Biometric capture device present, but Biometric Subtype entry was bypassed	Biometric = No	1
B4b8	Transaction exceeds floor limit	Floor Limit Checking = NO	1
B4b7	Lower consecutive offline limit exceeded	Velocity Checking = NO	1
B4b6	Upper consecutive offline limit exceeded	Velocity Checking = NO	1
B4b5	Transaction selected randomly for online processing	Random Transaction Selection = NO	1
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded	Biometric = No	1
B4b2	RFU or A selected Biometric Type not supported	Biometric = No	1
B4b1	XDA not failed	XDA = No	1
B5b8	Default TDOL used	Default TDOL = NO	1

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B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU	Always set	1
B5b3	CA ECC key missing	XDA AND ODE = No	1
B5b2	ECC key recovery failed	XDA AND ODE = No	1
B5b1	RFU	Always set	1

Note: even if not related to ICS-bitmask, the un-setting of RFU TVR bits is checked here.

TVR LT-bitmask

The LT-bitmask value is defined by the tool for a kernel configuration testing. This bitmask is depending on the LT setting used for a batch of test cases. LT-bitmask must be updated anytime the LT is changed:

- AIP
- AUC
- TDOL presence
- ATC value = 0
- Effective Date present
- Last Online ATC absent
- Lower Consecutive Limit and Upper Consecutive Limit absent
- …

The LT-bitmask is limited to the following TVR bits (any other TVR bit shall have the value '0' in this bitmask):

TVR	TVR Name	LT Data setting	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed	AIP States SDA is not supported	1
B1b6	ICC data missing	All Mandatory Data present in LT	1
B1b5	Card appears on terminal exception file		0
B1b4	DDA failed	AIP States DDA is not supported	1
B1b3	CDA failed	AIP States CDA is not supported	1
B1b2	SDA Selected	AIP States SDA is not supported	1
B1b1	XDA Selected	AIP States XDA is not supported	1
B2b8	ICC and terminal have different application versions	ICC application version not present in LT	1
B2b7	Expired application		0
B2b6	Application not yet effective	Effective Data not present in LT	1
B2b5	Requested service not allowed for card product	(AUC not present in LT) OR (AUC value = 'FFC0')	1
B2b4	New card	(ATC different from '0') OR (LOATC # '0') OR (LCOL AND UCOL not present)	1
B2b3	RFU		0

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B2b2	RFU or Biometric not performed or not successful		0
B2b1	RFU or Biometric template format supported		0
B3b8	Cardholder verification was not successful	AIP states Cardholder Verification is not supported	1
B3b7	Unrecognised CVM	(AIP states Cardholder Verification is not supported) OR (CVM list uses only EMVCo recognized Code)	1
B3b6	PIN Try Limit exceeded	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain Offline Enciphered PIN AND Plaintext PIN)	1
B3b5	PIN entry required and PIN pad not present or not working	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain any PIN)	1
B3b4	PIN entry required, PIN pad present, but PIN was not entered	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain any PIN)	1
B3b3	Online PIN entered	(AIP states Cardholder Verification is not supported) OR (CVM List does not contain Online PIN)	1
B3b2	RFU or Biometric required but Biometric capture device not working		0
B3b1	RFU or Biometric required, Biometric capture device present, but Biometric Subtype entry was bypassed		0
B4b8	Transaction exceeds floor limit		0
B4b7	Lower consecutive offline limit exceeded	Lower consecutive offline limit not present in LT	1
B4b6	Upper consecutive offline limit exceeded	Upper consecutive offline limit not present in LT	1
B4b5	Transaction selected randomly for online processing		0
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded		0
B4b2	RFU or A selected Biometric Type not supported		0
B4b1	XDA not Failed	AIP States XDA is not supported	1
B5b8	Default TDOL used	TDOL present in LT	1
B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU		0

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B5b3	RFU		0
B5b2	RFU		0
B5b1	RFU		0

TVR Transaction-bitmask

All bits of the Transaction-bitmask shall be set to zero for the time being.

TVR	TVR Name	LT Data setting	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed		0
B1b6	ICC data missing		0
B1b5	Card appears on terminal exception file		0
B1b4	DDA failed		0
B1b3	CDA failed		0
B1b2	SDA Selected		0
B1b1	XDA Selected		0
B2b8	ICC and terminal have different application versions		0
B2b7	Expired application		0
B2b6	Application not yet effective		0
B2b5	Requested service not allowed for card product		0
B2b4	New card		0
B2b3	RFU		0
B2b2	RFU or Biometric not performed or not successful		0
B2b1	RFU or Biometric template format supported		0
B3b8	Cardholder verification was not successful		0
B3b7	Unrecognised CVM		0
B3b6	PIN Try Limit exceeded		0
B3b5	PIN entry required and PIN pad not present or not working		0
B3b4	PIN entry required, PIN pad present, but PIN was not entered		0
B3b3	Online PIN entered		0
B3b2	RFU or Biometric required but Biometric capture device not working		0
B3b1	RFU or Biometric required, Biometric capture device present, but Biometric Subtype entry was bypassed		0
B4b8	Transaction exceeds floor limit		0
B4b7	Lower consecutive offline limit exceeded		0

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B4b6	Upper consecutive offline limit exceeded		0
B4b5	Transaction selected randomly for online processing		0
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded		0
B4b2	RFU or A selected Biometric Type not supported		0
B4b1	RFU		0
B5b8	Default TDOL used		0
B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU		0
B5b3	CA ECC key missing		0
B5b2	ECC key recovery failed		0
B5b1	RFU		0

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Terminal Settings

Several test cases require the use of predefined Terminal data setting listed in this section in one Terminal Setting named Termsetting1.

[Termsetting1]

Data	Tag (if any)	Value
Transaction Date	9A	Current Date
AID	**9F06**	**B000001234**
Application Version Number	9F09	3031
TAC-Default	-	0000000000
TAC-Denial	-	0000000000
TAC-Online	-	0000000000
AID	**9F06**	B000005678000000000000000 000000FF
TAC-Default	-	0000400000
TAC-Denial	-	0000000000
TAC-Online	-	0000400000
AID	**9F06**	**B000001235**
TAC-Default	-	2000000000
TAC-Denial	-	0000000000
TAC-Online	-	2000000000
AID	**9F06**	**B000001236**
TAC-Default	-	0080000000
TAC-Denial	-	0000000000
TAC-Online	-	0080000000
AID	**9F06**	**B000001237**
TAC-Default	-	0000400000
TAC-Denial	-	0000000000
TAC-Online	-	0000400000
AID	**9F06**	**B000001238**
TAC-Default	-	0040000000
TAC-Denial	-	0000000000
TAC-Online	-	0040000000
AID	**9F06**	**B000001239**
TAC-Default	-	0020000000
TAC-Denial	-	0000000000
TAC-Online	-	0020000000
AID	**9F06**	**B000001240**
TAC-Default	-	0040000000
TAC-Denial	-	0000000000
TAC-Online	-	0040000000

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AID	**9F06**	**B000001241**
TAC-Default	-	0000000000
TAC-Denial	-	0000000000
TAC-Online	-	0000000000
AID	**9F06**	**B000001242**
TAC-Default	-	0080000000
TAC-Denial	-	0000000000
TAC-Online	-	0080000000
AID	**9F06**	**B000001243**
TAC-Default	-	0060000000
TAC-Denial	-	0000000000
TAC-Online	-	0060000000
AID	**9F06**	**B000001244**
TAC-Default	-	0000000000
TAC-Denial	-	0000000000
TAC-Online	-	0000000000

Note: If the Kernel support less than 12 AIDs (as per ICS response), the above list will be limited to the first (in the order) AID listed above to reach the number of AID supported.
Note2: 6 is the minimum number of AID that the Kernel shall support.

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LT Settings

[LTSetting1] (updates only)

Data	Tag	Value
CDOL1	8C	95059B029F241D9F1906 9F25029F34039F3704 9F40059F33039A039F2103
CDOL2	8D	95059B029F241D9F1906 9F25029F34039F3704 9F40059F33039A039F21038A02

[LTSetting157] (differences with LTSetting1)

Data	Tag	Value
Track 1 Discretionary Data	9F1F	9F1F04C3C6

[LTSetting158] (differences with LTSetting1)

Data	Tag	Value
Track 1 Discretionary Data	9F1F	9F1F06CED7

[LTSetting159] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	000005DC0000000000060000

[LTSetting160] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	000005DC0000000000070000

[LTSetting161] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000005DC00080000

[LTSetting162] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000005DC00090000

[LTSetting163] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	000000000000000000040000

[LTSetting164] (differences with LTSetting1)

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Data	Tag	Value
CVM list	8E	00000000000000000000

[LTSetting165] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000002

[LTSetting166] (differences with LTSetting1)

Data	Tag	Value
AIP	82	2100

[LTSetting167] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000200

[LTSetting168] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	000003E8000000000206

[LTSetting169] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000003E80209

[LTSetting170] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	000003E8000000000207

[LTSetting171] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000003E80208

[LTSetting172] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000201

[LTSetting173] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000204

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[LTSetting174] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000205

[LTSetting175] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000202

[LTSetting176] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	000000000000000000410344030200

[LTSetting177] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	000000000000000041030200

[LTSetting178] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	000000000000000044030200

[LTSetting179] (differences with LTSetting1) :

Data	Tag	Value
Expiration Date	5F24	200101
IAC-Default	9F0D	0040000000

[LTSetting189] (differences with LTSetting1)

Data	Tag	Value
IAC-Denial	9F0E	00040000
IAC-Online	9F0F	00040000
IAC-Default	9F0D	00040000

[LTSetting190] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000102

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Section 7 – Test Cases Updates

New test cases include:

2CE.003.15	2CJ.087.06	2CJ.023.02	2CM.094.02	2CM.094.03

Revised test cases include:

2CA.041.00	2CE.003.00	2CJ.095.10	2CJ.178.00	2CS.016.03
2CA.041.04	2CJ.002.08	2CJ.095.11	2CJ.203.01	2CT.008.00
2CA.122.00	2CJ.018.00	2CJ.095.12	2CK.010.00	2CT.009.00
2CA.122.02	2CJ.018.01	2CJ.100.05	2CM.020.00	2CT.009.01
2CA.122.04	2CJ.018.02	2CJ.112.00	2CM.020.01	2CT.010.00
2CA.122.08	2CJ.018.03	2CJ.158.00	2CM.077.00	2CT.010.01
2CA.122.11	2CJ.018.04	2CJ.159.00	2CM.081.00	2CT.010.02
2CA.122.14	2CJ.018.09	2CJ.159.01	2CM.086.00	2CT.010.03
2CA.122.17	2CJ.018.10	2CJ.160.00	2CM.087.00	2CT.011.00
2CA.122.20	2CJ.074.00	2CJ.160.01	2CO.030.00	2CT.011.01
2CA.122.23	2CJ.084.01	2CJ.161.00	2CO.031.00	2CT.011.02
2CB.031.00	2CJ.087.00	2CJ.162.00	2CO.037.00	2CT.011.03
2CC.058.07	2CJ.088.01	2CJ.163.00	2CS.014.00	2CT.011.04
2CC.014.00	2CJ.089.00	2CJ.164.00	2CS.014.01	2CT.018.00
2CC.014.01	2CJ.089.02	2CJ.165.00	2CS.014.02	2CT.018.01
2CC.014.02	2CJ.089.04	2CJ.166.00	2CS.014.03	2CT.035.00
2CC.014.08	2CJ.089.06	2CJ.166.01	2CS.015.00	2CT.036.00
2CC.014.09	2CJ.089.08	2CJ.167.00	2CS.015.01	2CT.037.00
2CC.122.01	2CJ.089.10	2CJ.167.01	2CS.015.02	2CT.039.00
2CC.123.02	2CJ.095.06	2CJ.168.00	2CS.015.03	
2CC.134.06	2CJ.095.07	2CJ.168.01	2CS.015.04	
2CC.134.10	2CJ.095.08	2CJ.169.00	2CS.015.05	
2CC.134.12	2CJ.095.09	2CJ.169.02	2CS.016.00	

Deleted test cases include:

2CA.025.03	2CM.040.00	2CM.105.00	2CO.021.01	2CT.012.00
2CA.067.00	2CM.044.00	2CM.137.00	2CO.022.00	2CT.012.01
2CA.068.00	2CM.052.00	2CN.016.00	2CO.023.00	2CT.013.00
2CA.069.00	2CM.080.00	2CO.019.00	2CO.032.00	2CT.013.01
2CJ.177.00	2CM.082.00	2CO.020.00	2CO.033.00	
2CM.038.00	2CM.091.00	2CO.021.00	2CO.038.00	

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2CA.041.00 Coding of Bits and Bytes RFU

Revision: 44b

Objective: To ensure that the terminal sets to zeroes data (bits and bytes) indicated as RFU, unless otherwise stated. This applies particularly to TVR, TSI, Terminal capabilities, additional capabilities, GENERATE AC reference control parameter

Reference EMV 4.4: Book 3 - Section 6.3.6 - Coding of RFU Data
Book 3 - Section 6.5.5.2 - Command Message
Book 4 - Section A2 - Terminal Capabilities
Book 4 - Section A3 - Additional Terminal Capabilities

Terminal ICS Options: [NA]

Conditions: TVR, TSI, Terminal Capabilities and Additional Terminal Capabilities are requested in CDOL1

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Data received at first GENERATE AC:
  - RFU bits set to zero (TVR_B2b3='0').
  - RFU bits set to zero (TSI_B1b1='0').
  - RFU bits set to zero (TSI_B1b2='0').
  - RFU bits set to zero (TSI_B2b1='0').
  - RFU bits set to zero (TSI_B2b2='0').
  - RFU bits set to zero (TSI_B2b3='0').
  - RFU bits set to zero (TSI_B2b4='0').
  - RFU bits set to zero (TSI_B2b5='0').
  - RFU bits set to zero (TSI_B2b6='0').
  - RFU bits set to zero (TSI_B2b7='0').
  - RFU bits set to zero (TSI_B2b8='0').
  - RFU bits set to zero (TermCap_B1b1='0').
  - RFU bits set to zero (TermCap_B1b2='0').
  - RFU bits set to zero (TermCap_B1b3='0').
  - RFU bits set to zero (TermCap_B1b4='0').
  - RFU bits set to zero (TermCap_B1b5='0').
  - RFU bits set to zero (TermCap_B3b1='0').
  - RFU bits set to zero (TermCap_B3b2='0').

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2CA.041.00 Coding of Bits and Bytes RFU

- RFU bits set to zero (TermCap_B3b5=‘0’).
- RFU bits set to zero (AddTermCap_B2b1=‘0’).
- RFU bits set to zero (AddTermCap_B2b2=‘0’).
- RFU bits set to zero (AddTermCap_B2b3=‘0’).
- RFU bits set to zero (AddTermCap_B2b4=‘0’).
- RFU bits set to zero (AddTermCap_B2b5=‘0’).
- RFU bits set to zero (AddTermCap_B2b6=‘0’).
- RFU bits set to zero (AddTermCap_B2b7=‘0’).
- RFU bits set to zero (AddTermCap_B3b1=‘0’).
- RFU bits set to zero (AddTermCap_B3b2=‘0’).
- RFU bits set to zero (AddTermCap_B3b3=‘0’).
- RFU bits set to zero (AddTermCap_B3b4=‘0’).
- RFU bits set to zero (AddTermCap_B4b3=‘0’).
- RFU bits set to zero (AddTermCap_B4b4=‘0’).

• Parameter received at first GENERATE AC:

- Reference control parameter bits set to zero (P1_b1='0')
- Reference control parameter bits set to zero (P1_b2='0')
- Reference control parameter bits set to zero (P1_b3='0')
- Reference control parameter bits set to zero (P1_b6='0')

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2CA.041.04 Coding of Bits and Bytes RFU (5) (Implied)

Revision: 44b

Objective: To ensure that the terminal does not use the RFU bits, even when set to '1'. This applies to IAC

Reference EMV 4.4: Book 3 - Section 6.3.6 - Coding of RFU Data

Terminal ICS Options: [NA]

Conditions: IACs RFU bits of LT are set to '1'

Procedure:

- Transaction is performed with TermSetting1 and LTsetting189

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')

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2CA.122.00 Payment System reserved Data objects processing

Test No: **2CA.122.00** - *Revision* 44b - *Number of sub cases:* 7

Objective: To ensure that the terminal ignores Payment System reserved Data Objects when the selected AID is not an Payment System AID.

Reference EMV 4.4: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is B000001234
- Terminal AID Value is B000001234
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all reserved tags in the range '9F50' to '9F7F'.
- Case 01: FCI (tag 6F) of ADF contains all reserved tags in the range '9F50' to '9F6A' with values different from zero
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all reserved tags in the range '9F50' to '9F6A' with values different from zero.
- Case 03: Response to GENERATE AC commands contains all reserved tags in the range '9F50' to '9F7F' with values different from zero. and shall be encoded in Template 77.
- Case 04: Records contains all reserved tags in the range '9F50' to '9F7F' with values different from zero. (splitted over SFI 1 to 10).
- Case 05: Response to GET PROCESSING OPTIONS contains all reserved tags in the range '9F50' to '9F7F' with values different from zero. and shall be encoded in template 77
- Case 06: FCI (tag 6F) of ADF contains all reserved tags in the range '9F6B' to '9F7F' with values different from zero
- Case 07: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all reserved tags in the range '9F6B' to '9F7F' with values different from zero.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- Values of the reserved tags in the range '9F50' to '9F7F' received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.02 EMV reserved Data objects processing (1)

Test No: **2CA.122.02** – *Revision* 44b – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.4: Book 3 – Section Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book 3 - Section Annex A2 – Data Elements by tag

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is B000001234
- Terminal AID Value is B000001234
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in Book 3, Table 38 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags (reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).
- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the

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2CA.122.02 EMV reserved Data objects processing (1)

range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’(Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.04 Proprietary Data objects processing (1)

Test No: **2CA.122.04** - *Revision* 44b - *Number of sub cases:* 7

Objective: To ensure that the terminal ignores Proprietary Data Objects.

Reference EMV 4.4: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book 3 - Section Annex A2 – Data Elements by tag

Terminal Configuration (New format): [NA]

Conditions:

- All Data Elements listed in Book 3, Table 38 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally 50 proprietary tags including Tag DF09, DF10, DF11 and DF21 within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally 50 proprietary tags including Tag DF09, DF10, DF11 and DF21
- Case 03: Response to GET PROCESSING OPTIONS contains additionally 50 proprietary tags including Tag DF09, DF10, DF11 and DF21and shall be encoded in template 77
- Case 04: The PDOL of LT contains additionally 50 proprietary tags including Tag DF09, DF10, DF11 and DF21
- Case 05: The CDOL1 of LT contains additionally 50 proprietary tags including Tag DF09, DF10, DF11 and DF21
- Case 06: The CDOL2 of LT contains additionally 50 proprietary tags including Tag DF09, DF10, DF11 and DF21 and LT request ARQC on first Generate AC
- Case 07: The TDOL of LT contains additionally 50 proprietary tags including Tag DF09, DF10, DF11 and DF21

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC at first or second GENERATE AC.

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2CA.122.08 EMV reserved Data objects processing (5)

Test No: **2CA.122.08** – *Revision* 44b – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.4: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book 3 - Section Annex A2 – Data Elements by tag

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A000000025
- Terminal AID Value is A000000025
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in Book 3, Table 38 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags (reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).
- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

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2CA.122.08 EMV reserved Data objects processing (5)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.11 EMV reserved Data objects processing (8)

Test No: **2CA.122.11** – *Revision* 44b – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.4: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book 3 - Section Annex A2 – Data Elements by tag

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A0000000031010
- Terminal AID Value is VISA A0000000031010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F'.
- All Data Elements listed in Book 3, Table 38 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags (reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero (splitted over SFI 1 to 10).
- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero and shall be encoded in template 77

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2CA.122.11 EMV reserved Data objects processing (8)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F'received by the LT via the DOLs are set to "00' (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.14 EMV reserved Data objects processing (11)

Test No: **2CA.122.14** – *Revision* 44b – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.4: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book 3 - Section Annex A2 – Data Elements by tag

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A0000000041010
- Terminal AID Value is A0000000041010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in Book 3, Table 38 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags (reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).
- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

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2CA.122.14 EMV reserved Data objects processing (11)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.17 EMV reserved Data objects processing (14)

Test No: **2CA.122.17** – *Revision* 44b – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.4: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book 3 - Section Annex A2 – Data Elements by tag

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A0000000651010
- Terminal AID Value is A0000000651010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F'.
- All Data Elements listed in Book 3, Table 38 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags (reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero (splitted over SFI 1 to 10).
- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example '9F28'and '9F29', …) in the range '80' to '9E' and '9F01' to '9F4F' with values different from zero and shall be encoded in template 77

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2CA.122.17 EMV reserved Data objects processing (14)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.20 EMV reserved Data objects processing (17)

Test No: **2CA.122.20** – *Revision* 44b – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.4: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book 3 - Section Annex A2 – Data Elements by tag

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A000000333010101
- Terminal AID Value is A000000333010101
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in Book 3, Table 38 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags (reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).
- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

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2CA.122.20 EMV reserved Data objects processing (17)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CA.122.23 EMV reserved Data objects processing (20)

Test No: **2CA.122.23** – *Revision* 44b – *Number of sub cases:* 5

Objective: To ensure that the terminal ignores EMV reserved Data Objects (not used by EMV).

Reference EMV 4.4: Book 3 – Annex B1 Coding of the Tag Field of BER-TLV Data Objects

Book 3 - Section Annex A2 – Data Elements by tag

Terminal Configuration (New format): [NA]

Conditions:

- LT AID value is A0000001523010
- Terminal AID Value is A0000001523010
- PDOL, CDOL1, CDOL2 and TDOL of LT contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’.
- All Data Elements listed in Book 3, Table 38 are excluded from the below lists.
- Case 01: FCI (tag 6F) of ADF contains additionally all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero within the FCI template
- Case 02: FCI Issuer discretionary Data (tag BF0C) of ADF contains additionally all EMV primitive tags (reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero
- Case 03: Response to GENERATE AC contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77
- Case 04: Records contains: all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero (splitted over SFI 1 to 10).
- Case 05: Response to GET PROCESSING OPTIONS contains all EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’ with values different from zero and shall be encoded in template 77

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2CA.122.23 EMV reserved Data objects processing (20)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT until approval.

Pass Criteria:

- The terminal shall accept the card and process the transaction until completion, by requesting a TC either at First or Second GENERATE AC.
- Values of the EMV primitive tags reserved but not used in the EMV specification, (for example ‘9F28’and ‘9F29’, …) in the range ‘80’ to ‘9E’ and ‘9F01’ to ‘9F4F’received by the LT via the DOLs are set to “00’ (Note: for CDOL2 this is applicable only if a 2nd GENERATE AC has been performed)

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2CB.031.00 Application Selection Indicator: One Occurrence Permitted

Revision: 44b

Objective: To ensure that if the DFNAME in the ICC is longer than that of the AID in the terminal, but they are identical up to and including the last character in the terminal AID, the terminal shall check the Application Selection indicator.
If the indicator says only one occurrence is allowed the terminal does not add the AID to the candidate list and issue another SELECT command using the next AID.

Reference EMV 4.4: Book 1 - Section 12.3.3 - Using a List of AIDs

Terminal Configuration (New format): [NA]

Conditions:

- LT does not contain PSE
- DF Name 1 contained in FCI returned by the LT for first AID (AID1) selection is longer than that of terminal AID used for selection, but they are identical up to and including the last character in the terminal AID1.
- Terminal Application Selection Indicator only allows one occurrence for selected AID.
- LT shall contain another DF Name 2 exactly matching the terminal AID2.

Action: NA

Procedure: Application Selection using list of AIDs is performed.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- LT shall receive SELECT Command with AID name that exactly matches the ADF name from the LT.
- LT shall not receive SELECT Command with the longer DF Name 1
- LT shall not receive SELECT AID1 command with P2 set to 'Next occurrence
- After SELECT AID1 command, the LT shall receive another SELECT 'AID' command but with the AID2 name.
- .

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2CC.014.00 Retrieving CA Public Key used to Perform SDA: Key absent

Test No: **2CC.014.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the terminal supports Static Data Authentication, if no Certification Authority Public Keys is available for a given RID and Certification Authority Public Key Index, the terminal fails Static Data Authentication.

Reference EMV 4.4: Book 2 - Section 5.2 - Retrieval of Certification Authority Public Key

Terminal Configuration (New format): [SDA]

Conditions:

- Terminal does not contain the Certification Authority Public Key referenced in LT.
- AIP of LT indicates Static Data Authentication is supported (AIP byte 1 bit 7 = ‘1’).

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT (in particular Static Data Authentication).

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 7 = '1' (ie Offline Static Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.

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2CC.014.01 Retrieving CA Public Key used to Perform DDA: Key absent

Test No: **2CC.014.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the terminal supports Dynamic Data Authentication, and if no Certification Authority Public Keys is available for a given RID and Certification Authority Public Key Index, the terminal fails the Dynamic Data Authentication process.

Reference EMV 4.4: Book 2 - Section 6.2 - Retrieval of Certification Authority Public Key

Terminal Configuration (New format): [DDA]

Conditions:

- Terminal does not contain the Certification Authority Public Key referenced in LT.
- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 4 = '1' (ie Offline Dynamic Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.

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2CC.014.02 Retrieving CA Public Key used to Perform Combined DDA/AC: Key absent

Test No: **2CC.014.02** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the terminal supports Combined DDA/AC Data Authentication, and if no Certification Authority Public Keys is available for a given RID and Certification Authority Public Key Index, the terminal fails the Combined DDA/AC Data Authentication process.

Reference EMV 4.4: Book 2 - Section 6.2 - Retrieval of Certification Authority Public Key
Book 2 - Section 6.6 - Combined DDA/Application Cryptogram Generation (CDA)

Terminal Configuration (New format): [CDA] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- Terminal does not contain the Certification Authority Public Key referenced in LT.
- AIP of LT indicates CDA is supported (AIP byte 1 bit 1 = ‘1’).
- IAC’s and TAC’s are set so that TC is requested at first Gen AC.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The Terminal shall not request CDA for the GENERATE AC
- The terminal shall process the transaction until completion, by requesting a TC according to TAC and IAC setting.
- TVR byte 1, bit 3 = '1' (ie Offline Combined DDA/AC Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.

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2CC.014.08 Retrieving CA Public Key used to Perform Combined DDA/AC: Key absent (9)

Test No: **2CC.014.08** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the terminal supports Combined DDA/AC Data Authentication, and if no Certification Authority Public Keys is available for a given RID and Certification Authority Public Key Index, the terminal fails the Combined DDA/AC Data Authentication process.

Reference EMV 4.4: Book 2 - Section 6.2 - Retrieval of Certification Authority Public Key
Book 2 - Section 6.6 - Combined DDA/Application Cryptogram Generation (CDA)

Terminal Configuration (New format): [CDA] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- Terminal does not contain the Certification Authority Public Key referenced in LT.
- AIP of LT indicates CDA is supported (AIP byte 1 bit 1 = ‘1’).
- IAC’s and TAC’s are set so that ARQC is requested at first Gen AC.
- Transaction is online approved.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The Terminal shall not request CDA for the 1st and 2nd GENERATE AC
- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TVR byte 1, bit 3 = '1' (ie Offline Combined DDA/AC Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.

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2CC.014.09 Retrieving CA Public Key used to Perform Combined DDA/AC: Key absent (10)

Test No: **2CC.014.09** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the terminal supports Combined DDA/AC Data Authentication, and if no Certification Authority Public Keys is available for a given RID and Certification Authority Public Key Index, the terminal fails the Combined DDA/AC Data Authentication process.

Reference EMV 4.4: Book 2 - Section 6.2 - Retrieval of Certification Authority Public Key
Book 2 - Section 6.6 - Combined DDA/Application Cryptogram Generation (CDA)

Terminal Configuration (New format): [CDA] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- Terminal does not contain the Certification Authority Public Key referenced in LT.
- AIP of LT indicates CDA is supported (AIP byte 1 bit 1 = ‘1’).
- IAC’s and TAC’s are set so that ARQC is requested at first Gen AC.
- Transaction is online declined.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The Terminal shall not request CDA for the 1st and 2nd GENERATE AC
- The terminal shall process the transaction until completion, by requesting an AAC at second GENERATE AC.
- TVR byte 1, bit 3 = '1' (ie Offline Combined DDA/AC Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.

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2CC.058.07 Certification Revocation List update, addition (3)

Revision: 44b

Objective: To ensure that the terminal is be able to update the Certification Revocation List by adding an entry.

Reference EMV 4.4: Book 2 - Section 5.1.2 - Certification Revocation List

Terminal ICS Options: [CDA] AND [Key Revocation]

Conditions:

- Pre-Conditions: Test 2CC.058.06 has to be run before
- LTsetting143:
  - AIP of LT indicates CDA is supported but no other authentication methods
- Action:
  - CRL entry 'xx' is removed and replaced by CRL entry 'yy' from the CRL list before undertaking an EMV transaction. A valid CRL entry is loaded to the device, where the LT Issuer Public Key Certificate in LT is calculated with RID, CA Public Key Index and Certificate Serial Number to correspond to this valid entry. A default acquirer process as documented by the device vendor is performed to update the CRL

Procedure:

- Transaction is performed with TermSetting1 and LTsetting143

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Data received in Kernel Data Output:
  - CDA failed (TVR_B1b3='1').

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2CC.122.01 Response to GENERATE AC when responding AAC is in format 1 or 2 (2)

Revision: 44b

Objective: To ensure that the terminal can use the ICC response format 1 or 2 (as specified in book 3 part I) in Enhanced Combined DDA/AC, when LT responds AAC to GENERATE AC command.

Reference EMV 4.4: Book 2 - Section 6.6.1 - Dynamic Signature Generation

Terminal ICS Options: [CDA] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- LTsetting1:
  - AIP indicates CDA is supported
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’) and command response is in format 2.
- Case 01
  - LT_response at second GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an AAC (‘00’) and command response is in format 1.
- Case 02
  - LT_response at second GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an AAC (‘00’) and command response is in format 2.

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - SDA not used (TVR_B1b7=‘0’).
  - DDA not used (TVR_B1b4=‘0’).
- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TSI_B1b8=‘1’).

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2CC.123.02 Signed Dynamic Application Data Length (3)

Revision: 44b

Objective: To ensure that the terminal compares the Signed Dynamic Application Data length with the ICC Public Key length for Combined DDA/AC Data Authentication.

Reference EMV 4.4: Book 2 - Section 6.6.2 - Dynamic Signature Verification

Terminal ICS Options: [Online Only] AND [Normal Default Action Code processing] AND ([CDA Mode1] OR [CDA Mode2])

Conditions:

- LTsetting1:
  - AIP indicates CDA is supported
- LT_response at second GENERATE AC
  - Length of Signed Dynamic Application Data is different from the ICC Public Key length
- Issuer_response:
  - The issuer does not respond to online processing request. Terminal unable to go Online.

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Data received at first GENERATE AC:
  - SDA not used (TVR_B1b7=‘0’).
  - DDA not used (TVR_B1b4=‘0’).
- Terminal display: 'Declined' message
- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TSI_B1b8=‘1’).
  - Combined Data Authentication failed (TVR_B1b3=‘1’).

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2CC.134.06 Response to GENERATE AC when responding TC or ARQC is not in format 1 (Implied) (2)

Revision: 44b

Objective: To ensure that the terminal does not use the ICC response format 1 in Enhanced Combined DDA/AC (as specified in book 3 part I) when responding TC or ARQC.

Reference EMV 4.4: Book 2 - Section 6.6.1 - Dynamic Signature Generation

Terminal ICS Options: [Online Only] AND [CDA Mode1]

Conditions:

- LTsetting107:
  - AIP indicates CDA is supported
- Case 01
  - LT_response at first GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’) in format 1
- Case 02
  - LT_response at first GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’) in format 2
  - LT_response at second GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an TC (‘40’) in format 1

Procedure:

- Transaction is performed with TermSetting1 and LTsetting107

Pass Criteria:

- Terminal display: ‘Declined’ OR 'Terminated' message

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2CC.134.10 Response to GENERATE AC when responding TC or ARQC is not in format 2 (Implied) (3)

Revision: 44b

Objective: To ensure that the terminal does not use the ICC response format 1 in Enhanced Combined DDA/AC (as specified in book 3 part I) when responding TC or ARQC.

Reference EMV 4.4: Book 2 - Section 6.6.1 - Dynamic Signature Generation

Terminal ICS Options: [Online Only] AND [Normal Default Action Code processing] AND ([CDA Mode1] OR [CDA Mode2])

Conditions:

- LTsetting107:
  - AIP indicates CDA is supported
- Case 01
  - LT_response at first GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’) in format 1
- Case 02
  - LT_response at first GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’) in format 2
  - LT_response at second GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an TC (‘40’) in format 1
  - Issuer_response:
    - The Issuer does not respond to online processing request. Terminal unable to go Online.

Procedure:

- Transaction is performed with TermSetting1 and LTsetting107

Pass Criteria:

- Terminal display: ‘Declined’ OR 'Terminated' message

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2CC.134.12 Response to GENERATE AC when responding TC or ARQC is not in format 1 (Implied) (3)

Revision: 44b

Objective: To ensure that the terminal does not use the ICC response format 1 in Enhanced Combined DDA/AC (as specified in book 3 part I) when responding TC or ARQC.

Reference EMV 4.4: Book 2 - Section 6.6.1 - Dynamic Signature Generation

Terminal ICS Options: [Online Only] AND ([CDA Mode1] OR [CDA Mode4])

Conditions:

- LTsetting107:
  - AIP indicates CDA is supported
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’) in format 2
- LT_response at second GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an TC (‘40’) in format 1

Procedure:

- Transaction is performed with TermSetting1 and LTsetting107

Pass Criteria:

- Terminal display: ‘Declined’ OR 'Terminated' message

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2CE.003.00 Recognition of ‘Space’ characters in AN formatted Data Objects

Test No: **2CE.003.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that terminal accepts data objects from the ICC with ‘space’ characters included in AN defined fields.

Reference EMV 4.4: Book 3 - Section 43 - Data Element Format Conventions

Terminal Configuration (New format): [Display] AND [Cardholder Confirmation]

Conditions:

- LT shall contain the following data objects with at least one ‘space’ character in each data object:
  - Application Preferred Name.
  - Application Label.
- LT shall contain the Issuer Code Table Index.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- The terminal shall display either the ‘Application Preferred Name’ or the ‘Application Label both with the ‘space’ character present

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2CE.003.15 Acceptance of format errors on data objects (2)

Revision: 44b

Objective: The terminal accepts specific data objects from the ICC with format errors, when processing the transaction

Reference EMV 4.4: Book 3 - Section 7.5 – Erroneous or Missing Data in the ICC

Terminal ICS Options: [NA]

Conditions:

- Case 01
  - LTsetting157:
    - LT contains the Track 1 Discretionary Data with value: 9F1F04C3C6
- Case 02
  - LTsetting158:
    - LT contains the Track 1 Discretionary Data with value: 9F1F06CED7

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting157
- Case 02: Transaction is performed with TermSetting1 and LTsetting158

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')

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2CJ.002.08 All bits in TSI and TVR are set to 0b (9)

Revision: 44b

Objective: To ensure that all TVR and TSI bits are set to 0b when the terminal starts the transaction.

Reference EMV 4.4: Book 3 - Section 10.1 - Initiate Application Processing

Terminal ICS Options: [Offline/Online Capable] OR [Online Only]

Conditions:

- Transaction 01: A first transaction is performed raising the TVR bits: Byte 2 bit 8 to 5 (ICC have different application version, Expired Application, Application not yet effective, Requested service not allowed for card product), Byte 3 bit 8 to 7 (Cardholder verification not successful, Unrecognised CVM) and Byte 5 bit 7 to 5 (Issuer Authentication failed, Script processing failed before final GENERATE AC, Script processing failed after final GENERATE AC) and raising the TSI bits: Byte 1 bit 7 to 5, 3 (Cardholder verification was performed, Card risk management was performed, Issuer Authentication was performed, script processing was performed.
  - TermSetting1:
    - Transaction Date set between 200101 and 480101 (current local date of testing)
  - LTsetting108:
    - Expiration Date has the value '200101'
    - Effective Date has the value '480101'
    - Application Version Number has the value '3032'
    - Application Usage Control has the value '0000'
    - Issuer Country Code is returned
    - CVM List has the value '00000000000000003F00'
    - IAC-Online has the value '0040000000'
  - LT_response at All Script commands
    - SW1SW2: 6984
  - LT_response at External Authenticate command
    - SW1SW2: 6300
  - Issuer_response:
    - Issuer Script '71' contain one Issuer Script Command
    - Issuer Script '72' contain one Issuer Script Command
- Transaction 02: A second transaction is performed correctly and the PDOL request the TVR and TSI
  - LTsetting1:

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2CJ.002.08 All bits in TSI and TVR are set to 0b (9)

- PDOL is present in LT and request the TVR and TSI

Procedure:

- Transaction 01: Transaction is performed with TermSetting1 and LTsetting108
- Transaction 02: Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Transaction 01:
  - Data received at first GENERATE AC:
    - ICC and terminal have different application versions (TVR_B2b8=‘1’)
    - Expired application (TVR_B2b7=‘1’)
    - Application not yet effective (TVR_B2b6=‘1’)
    - Requested service not allowed for card product (TVR_B2b5=‘1’)
    - Cardholder verification was not successful (TVR_B3b8=‘1’)
    - Unrecognised CVM (TVR_B3b7=‘1’)
    - Cardholder verification was performed (TSI_B1b7='1')
  - Data received at second GENERATE AC:
    - Card risk management was performed (TSI_B1b6='1')
    - Script processing failed before final GENERATE AC (TVR_B5b6=‘1’)
    - Script processing was performed (TSI_B1b3='1')
    - Issuer Authentication Failed (TVR_B5b7=‘1’)
  - Data received in Kernel Data Output:
    - Transaction is approved with a TC: (CID=’40’)
    - Script processing failed after final GENERATE AC (TVR_B5b5=‘1’)
- Transaction 02:
  - Data received at Get Processing Options:
    - All TVR bits are set to zero
    - All TSI bits are set to zero

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2CJ.018.00 Record Data Format: Proprietary data participating in offline data authentication

Test No: **2CJ.018.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal is able to read and include in Offline Static Data authentication, data objects located in proprietary files, provided that proprietary files are TLV-coded.

Reference EMV 4.4: Book 3 - Section 10.3 - Offline Data Authentication

Terminal Configuration (New format): [SDA]

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file, listed in AFL, and included in the data to be signed.
- AIP of LT indicates Static Data Authentication is supported (AIP byte 1 bit 7 = ‘1’).
- EMV Data Object located in proprietary files will be TLV coded with record tag ‘70’.
- Signed Static Application Data is valid, including in the computation the tag '70' and associated length of the record contained in the proprietary files.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT (in particular Static Data Authentication).

Pass Criteria:

- The terminal shall process the transaction until completion, with an approval (either at 1st GEN AC or 2nd GEN AC).
- TVR byte 1, bit 7 = '0' (ie Offline Static Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '1' (ie Offline Static Data Authentication selected) received at 1st GENERATE AC.

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2CJ.018.01 Record Data Format: Proprietary data participating in offline data authentication (2)

Test No: **2CJ.018.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal is able to read and include in Offline Dynamic Data Authentication, data objects located in proprietary files, provided that proprietary files are TLV-coded.

Reference EMV 4.4: Book 3 - Section 10.3 - Offline Data Authentication

Terminal Configuration (New format): [DDA]

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file, listed in AFL, and included in the data to be signed.
- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).
- EMV Data Object located in proprietary files will be TLV coded with record tag ‘70’.
- ICC Public Key Certificate is valid including in the computation the tag '70' and associated length of the record contained in the proprietary files.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT (in particular Dynamic Data Authentication).

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.

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2CJ.018.02 Record Data Format: Non-TLV coded proprietary data participating in offline data authentication (2)

Test No: **2CJ.018.02** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal fails Dynamic Data Authentication when data objects located in proprietary files are not TLV coded with record tag ‘70’.

Reference EMV 4.4: Book 3 - Section 10.3 - Offline Data Authentication

Terminal Configuration (New format): [DDA]

Conditions:

- An EMV Data Object is included in a record, located in a proprietary (SFI 11 to 30) file, listed in AFL, and included in the data to be signed.
- EMV Data Object located in proprietary files is not TLV coded with record tag ‘70’.
- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).
- ICC Public Key Certificate is valid.
- Signed Static Application Data is good.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 4 = ‘1’ (ie Offline Dynamic Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = ‘1’ (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.

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2CJ.018.03 Record Data Format: Non-TLV coded proprietary data participating in offline data authentication (3)

Test No: **2CJ.018.03** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal fails Combined DDA/AC Data Authentication when data objects located in proprietary files are not TLV coded with record tag ‘70’.

Reference EMV 4.4: Book 3 - Section 10.3 - Offline Data Authentication
Book 2 - Section 6.6 - Combined DDA/Application Cryptogram Generation (CDA)

Terminal Configuration (New format): [CDA] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file (SFI 11 to 30), listed in AFL, and included in the data to be signed.
- EMV Data Object located in proprietary files is not TLV coded with record tag ‘70’.
- AIP of LT indicates CDA is supported (AIP byte 1 bit 1 = ‘1’).
- IAC’s and TAC’s are set so that TC is requested at first Gen AC.
- ICC Public Key Certificate is valid.
- Signed Static Application Data is good

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The Terminal shall not request CDA for the GENERATE AC
- The terminal shall process the transaction until completion, by requesting a TC at first GENERATE AC.
- TVR byte 1, bit 3 = ‘1’ (e Offline Combined DDA/AC Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = ‘1’ (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.

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2CJ.018.04 Record Data Format: Proprietary data participating in offline data authentication (3)

Test No: **2CJ.018.04** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal is able to read and include in Offline combined DDA/AC data authentication, data objects located in proprietary files, provided that proprietary files are TLV-coded.

Reference EMV 4.4: Book 3 - Section 10.3 - Offline Data Authentication

Terminal Configuration (New format): [CDA]

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file, listed in AFL, and included in the data to be signed.
- AIP of LT indicates CDA is supported (AIP byte 1 bit 1 = ‘1’).
- EMV Data Object located in proprietary files will be TLV coded with record tag ‘70’.
- ICC Public Key Certificate is valid including in the computation the tag '70' and associated length of the record contained in the proprietary files.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Offline Combined DDA/AC Data Authentication succeeded) contained in Kernel Data Output.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) contained in Kernel Data Output (this pass criteria apply only if CDA is requested).
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.

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2CJ.018.09 Record Data Format: Non-TLV coded proprietary data participating in offline data authentication (10)

Test No: **2CJ.018.09** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal fails Combined DDA/AC Data Authentication when data objects located in proprietary files are not TLV coded with record tag ‘70’.

Reference EMV 4.4: Book 2 - Section 6.6 - Combined DDA/Application Cryptogram Generation (CDA)
Book 3 - Section 10.3 - Offline Data Authentication

Terminal Configuration (New format): [CDA] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file (SFI 11 to 30), listed in AFL, and included in the data to be signed.
- EMV Data Object located in proprietary files is not TLV coded with record tag ‘70’.
- AIP of LT indicates CDA is supported (AIP byte 1 bit 1 = ‘1’).
- IAC’s and TAC’s are set so that ARQC is requested at first Gen AC.
- Transaction is online approved.
- ICC Public Key Certificate is valid.
- Signed Static Application Data is good

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The Terminal shall not request CDA for the 1st and 2nd GENERATE AC
- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TVR byte 1, bit 3 = ‘1’ (e Offline Combined DDA/AC Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = ‘1’ (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.

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2CJ.018.10 Record Data Format: Non-TLV coded proprietary data participating in offline data authentication (11)

Test No: **2CJ.018.10** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal fails Combined DDA/AC Data Authentication when data objects located in proprietary files are not TLV coded with record tag ‘70’.

Reference EMV 4.4: Book 2 - Section 6.6 - Combined DDA/Application Cryptogram Generation (CDA)
Book 3 - Section 10.3 - Offline Data Authentication

Terminal Configuration (New format): [CDA] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file (SFI 11 to 30), listed in AFL, and included in the data to be signed.
- EMV Data Object located in proprietary files is not TLV coded with record tag ‘70’.
- AIP of LT indicates CDA is supported (AIP byte 1 bit 1 = ‘1’).
- IAC’s and TAC’s are set so that ARQC is requested at first Gen AC.
- Transaction is online declined
- ICC Public Key Certificate is valid.
- Signed Static Application Data is good

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The Terminal shall not request CDA for the 1st and 2nd GENERATE AC
- The terminal shall process the transaction until completion, by requesting an AAC at second GENERATE AC.
- TVR byte 1, bit 3 = ‘1’ (e Offline Combined DDA/AC Data Authentication failed) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = ‘1’ (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.

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2CJ.023.02 Both the Offline DDA and the Offline SDA are supported (3)

Revision: 44b

Objective: To ensure that if both the card and the terminal support Offline Static Data Authentication and Offline Dynamic Data Authentication, and either the card or terminal (or both) does not support Enhanced Combined DDA/AC Generation, the terminal performs only Offline Dynamic Data Authentication.

Reference EMV 4.4: Book 3 - Section 10.3 - Offline Data Authentication

Terminal ICS Options: [DDA] AND NOT[CDA]

Conditions:

- LTsetting166:
  - AIP indicates DDA and CDA only are supported

Procedure:

- Transaction is performed with TermSetting1 and LTsetting166

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Data received at first GENERATE AC:
  - SDA not used (TVR_B1b7='0').
  - DDA is successful (TVR_B1b4='0').
  - CDA not used (TVR_B1b3='0').
  - SDA not selected (TVR_B1b2='0').
  - Offline Data Authentication was performed (TSI_B1b8='1').

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2CJ.074.00 Supported CVR condition: If Terminal supports the CVM and CVM is online PIN

Revision: 44b

Objective: To ensure that the terminal performs the CVM if the condition code is satisfied and CVM code is fail CVM.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Terminal ICS Options: [Online Enciphered PIN]

Conditions:

- LTsetting107:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000203', 'Enciphered Online PIN verification, if Terminal supports the CVM' (02 03)
- Action:
  - Cardholder enters a valid PIN.

Procedure: Transaction is performed with TermSetting1 and LTsetting107

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 03 00'

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2CJ.084.01 ICC Data required by the CVM Condition Code is missing (Amount Authorised)

Revision: 44b

Objective: To ensure that if the ICC data required by the condition expressed in the second byte of a Cardholder Verification Rule is not present, the terminal bypasses the rules and proceeded to the next rule.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Terminal ICS Options: NOT[Amount known before CVM Processing]

Conditions:

- Application Currency Code is present in the LT.
- Transaction Currency Code is equal to Application Currency Code.
- Case 01
  - Transaction amount is under X.
  - LTsetting159:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '000005DC0000000000060000', 'Fail CVM if transaction is in the application currency and is under X' value (00 06) followed by 'Fail CVM, always' (00 00).
- Case 02
  - Transaction amount is over X.
  - LTsetting160:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '000005DC0000000000070000', 'Fail CVM if transaction is in the application currency and is over X' value (00 07) followed by 'Fail CVM, always' (00 00).
- Case 03
  - Transaction amount is under Y.
  - LTsetting161:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '00000000000005DC00080000', 'Fail CVM if transaction is in the application currency and is under Y' value (00 08) followed by 'Fail CVM, always' (00 00).
- Case 04
  - Transaction amount is over Y.
  - LTsetting162:

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2CJ.084.01 ICC Data required by the CVM Condition Code is missing (Amount Authorised)

- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM List has the value '00000000000005DC00090000', 'Fail CVM if transaction is in the application currency and is over Y' value (00 09) followed by 'Fail CVM, always' (00 00).

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting159
- Case 02: Transaction is performed with TermSetting1 and LTsetting160
- Case 03: Transaction is performed with TermSetting1 and LTsetting161
- Case 04: Transaction is performed with TermSetting1 and LTsetting162

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '00 00 01'

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2CJ.087.00 Second byte of Cardholder Verification Rule satisfied and CVM Code is Fail CVM (1)

Revision: 44b

Objective: To ensure that the terminal performs the CVM if the condition code is satisfied and CVM code is fail CVM.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification
Book 4 – Section 6.3.4 - Cardholder Verification Processingn

Terminal ICS Options: [NA]

Conditions:

- LTsetting164:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000000', 'Fail CVM, always' (00 00)

Procedure: Transaction is performed with TermSetting1 and LTsetting164

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8=‘1’)
  - Recognized CVM (TVR_B3b7=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '00 00 01'

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2CJ.087.06 Second byte of Cardholder Verification Rule satisfied and CVM Code is Fail CVM (1)

Revision: 44b

Objective: To ensure that the terminal performs the CVM if the condition code is satisfied and CVM code is fail CVM.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification
Book 4 – Section 6.3.4 - Cardholder Verification Processingn

Terminal ICS Options: [Goods] OR [Services] OR [Inquiry] OR [Transfer] OR [Payment] OR [Administrative] OR [CashDeposit]

Conditions:

- Transaction Type is not cash and not purchase with cashback
- LTsetting165:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000002', 'not unattended cash and not manual cash, and not purchase with cashback (00 02).

Procedure: Transaction is performed with TermSetting1 and LTsetting165

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8=‘1’)
  - Recognized CVM (TVR_B3b7=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '00 02 01'

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2CJ.088.01 Second byte of CVR satisfied and CVM Code is Plain text PIN verified by ICC (Plain text PIN verified by ICC not supported)

Revision: 44b

Objective: To ensure that if the terminal does not support Plaintext PIN verified by ICC, the terminal performs the CVM if the condition code is satisfied and CVM code is Plaintext PIN verified by ICC.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification, Book 4 – Section 6.3.4 - Cardholder Verification Processing

Terminal ICS Options: NOT[Offline Plaintext PIN] AND ([Goods] OR [Services] OR [Inquiry] OR [Transfer] OR [Payment] OR [Administrative] OR [CashDeposit])

Conditions:

- LTsetting190:
  - AIP of LT indicates Cardholder Verification is supported
  - CVM List has the value '00000000000000000102' (Plaintext PIN verified by ICC, If not unattended cash and not manual cash, and not purchase with cashback).

Procedure:

- Transaction is performed with TermSetting1 and LTsetting190

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC or (P1_b8='0' AND P1_b7='1')
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8='1')
  - Recognised CVM (TVR_B3b7='0')
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '3F 00 01'

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2CJ.089.00 Second byte of CVR satisfied and CVM Code is - Enciphered PIN Online (Enciphered PIN Online supported)

Revision: 44b

Objective: To ensure that if terminal supports Enciphered PIN Online, the terminal performs the CVM if the condition code is satisfied and CVM code is Enciphered PIN Online.
To ensure that the terminal sets the CVM Results byte 3 to 'failed' when the last CVM performed was not considered as successful.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification
Book 4 - Section 6.3.4.5 - CVM Results

Terminal ICS Options: [Online Enciphered PIN]

Conditions:

- LTsetting167:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000200', 'Enciphered Online PIN verification, Always' (02 00)
- Action:
  - Cardholder enters a valid PIN.

Procedure: Transaction is performed with TermSetting1 and LTsetting167

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 00 00'

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2CJ.089.02 Second byte of CVR satisfied and CVM Code is - Enciphered PIN Online (Enciphered PIN Online supported) (2)

Revision: 44b

Objective: To ensure that if terminal supports Enciphered PIN Online, the terminal performs the CVM if the condition code is satisfied and CVM code is Enciphered PIN Online

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Terminal ICS Options: [Online Enciphered PIN] AND [Amount known before CVM Processing]

Conditions:

- Case 01
  - Transaction amount is under X
  - LTsetting168:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '000003E8000000000206', 'Enciphered Online PIN verification, transaction is in the application currency and is under X' value (02 06).
- Case 02
  - Transaction amount is over Y
  - LTsetting169:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '00000000000003E80209', 'Enciphered Online PIN verification, transaction is in the application currency and is over Y' value (02 09).
- Case 03
  - Transaction amount is over X
  - LTsetting170:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '000003E8000000000207', 'Enciphered Online PIN verification, transaction is in the application currency and is over X' value (02 07).
- Case 04
  - Transaction amount is under Y
  - LTsetting171:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).

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2CJ.089.02 Second byte of CVR satisfied and CVM Code is - Enciphered PIN Online (Enciphered PIN Online supported) (2)

- CVM List has the value '00000000000003E80208', 'Enciphered Online PIN verification, transaction is in the application currency and is under Y' value (02 08).

Procedure:

- Action:
  - Cardholder enters a valid PIN.
- Case 01: Transaction is performed with TermSetting1 and LTsetting168
- Case 02: Transaction is performed with TermSetting1 and LTsetting169
- Case 03: Transaction is performed with TermSetting1 and LTsetting170
- Case 04: Transaction is performed with TermSetting1 and LTsetting171

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8='0')
  - Online PIN entered (TVR_B3b3='1')
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 xx 00' where xx = number of the CVM condition of the sub case in question.

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2CJ.089.04 Second byte of CVR satisfied and CVM Code is - Enciphered PIN Online (Enciphered PIN Online supported) (3)

Revision: 44b

Objective: To ensure that if terminal supports Enciphered PIN Online, the terminal performs the CVM if the condition code is satisfied and CVM code is Enciphered PIN Online
To ensure that the terminal sets the CVM Results byte 3 to 'failed' when the last CVM performed was not considered as successful.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification
Book 4 - Section 6.3.4.5 - CVM Results

Terminal ICS Options: [Online Enciphered PIN] AND [Cash] AND [Unattended]

Conditions:

- Transaction type is cash.
- LTsetting172:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000201', 'Enciphered Online PIN verification, unattended cash' (02 01).
- Action:
  - Cardholder enters a valid PIN.

Procedure:

Transaction is performed with TermSetting1 and LTsetting172

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 01 00'.

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2CJ.089.06 Second byte of CVR satisfied and CVM Code is - Enciphered PIN Online (Enciphered PIN Online supported) (4)

Revision: 44b

Objective: To ensure that if terminal supports Enciphered PIN Online, the terminal performs the CVM if the condition code is satisfied and CVM code is Enciphered PIN Online.
To ensure that the terminal sets the CVM Results byte 3 to 'failed' when the last CVM performed was not considered as successful.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification
Book 4 - Section 6.3.4.5 - CVM Results

Terminal ICS Options: [Online Enciphered PIN] AND [Attended] AND [Cash]

Conditions:

- Transaction type is cash.
- LTsetting173:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000204', 'Enciphered Online PIN verification, manual cash' (02 04).
- Action:
  - Cardholder enters a valid PIN.

Procedure:

Transaction is performed with TermSetting1 and LTsetting173

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 04 00'.

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2CJ.089.08 Second byte of CVR satisfied and CVM Code is - Enciphered PIN Online (Enciphered PIN Online supported) (5)

Revision: 44b

Objective: To ensure that if terminal supports Enciphered PIN Online, the terminal performs the CVM if the condition code is satisfied and CVM code is Enciphered PIN Online.
To ensure that the terminal sets the CVM Results byte 3 to 'failed' when the last CVM performed was not considered as successful.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification
Book 4 - Section 6.3.4.5 - CVM Results

Terminal ICS Options: [Online Enciphered PIN] AND [Cashback]

Conditions:

- Transaction type is purchase with cashback.
- LTsetting174:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000205', 'Enciphered Online PIN verification, manual cash' (02 05).
- Action:
  - Cardholder enters a valid PIN.

Procedure:

Transaction is performed with TermSetting1 and LTsetting174

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 05 00'.

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2CJ.089.10 Second byte of CVR satisfied and CVM Code is - Enciphered PIN Online (Enciphered PIN Online supported) (6)

Revision: 44b

Objective: To ensure that if terminal supports Enciphered PIN Online, the terminal performs the CVM if the condition code is satisfied and CVM code is Enciphered PIN Online.
To ensure that the terminal sets the CVM Results byte 3 to 'failed' when the last CVM performed was not considered as successful.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification
Book 4 - Section 6.3.4.5 - CVM Results

Terminal ICS Options: [Online Enciphered PIN] AND ([Goods] OR [Services] OR [Inquiry] OR [Transfer] OR [Payment] OR [Administrative] OR [CashDeposit])

Conditions:

- Transaction type is is not cash and not purchase with cashback.
- LTsetting175:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000202', 'Enciphered Online PIN verification, 'not unattended cash and not manual cash, and not purchase with cashback' (02 02).
- Action:
  - Cardholder enters a valid PIN.

Procedure:

Transaction is performed with TermSetting1 and LTsetting175

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 02 00'.

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2CJ.095.06 Second byte of CVR satisfied and CVM Code not understood by the terminal (7)

Test No: **2CJ.095.06** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure that the terminal sets the 'Unrecognized CVM ' bit in the TVR to 1b, if the condition code is satisfied and CVM code is not understood by the terminal (RFU)

Reference Book 3 - Section 10.5 - Cardholder Verification

EMV 4.4: Book 4 – Section 6.3.4 - Cardholder Verification Processing

Terminal Configuration (New format): NOT[Biometric]

Conditions:

- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM List is 'RFU' with the following conditions satisfied:
- Case 01: condition is 'always' (3F 00)
- Case 02: CVM List is 'RFU, always' (07 00)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '1' (ie Cardholder verification was not successful
- TVR byte 3, bit 7 = '1' (ie Unrecognized CVM) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder Verification was performed) received at 1st GENERATE AC.
- CVM Result = 3F 00 01

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2CJ.095.07 Second byte of CVR satisfied and CVM Code not understood by the terminal (8)

Test No: **2CJ.095.07** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure that the terminal sets the 'Unrecognized CVM ' bit in the TVR to 1b, if the condition code is satisfied and CVM code is not understood by the terminal (RFU)

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification
Book 4 – Section 6.3.4 - Cardholder Verification Processing

Terminal Configuration (New format): NOT[Biometric] AND [Amount known before CVM Processing]

Conditions:

- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- Transaction Amount is known before CVM Processing.
- Transaction Amount is in the Application Currency
- CVM List is 'RFU' with the following conditions satisfied:
- Case 01: condition is 'transaction is under X (and Amount is under X)' (07 06)
- Case 02: condition is 'transaction is over Y (and Amount is over Y)' (07 09)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 7 = '1' (ie Unrecognized CVM) received at 1st GENERATE AC.
- TVR byte 3, bit 8 = '1' (ie Cardholder verification failed) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder Verification was performed) received at 1st GENERATE AC.
- CVM Result = 3F 00 01

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2CJ.095.08 Second byte of CVR satisfied and CVM Code not understood by the terminal (9)

Test No: **2CJ.095.08** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal sets the 'Unrecognized CVM ' bit in the TVR to 1b, if the condition code is satisfied and CVM code is not understood by the terminal (RFU)

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Book 4 – Section 6.3.4 - Cardholder Verification Processing

Terminal Configuration (New format): NOT[Biometric] AND [Unattended] AND [Cash]

Conditions:

- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM List is 'RFU' with the following conditions satisfied:
- condition is 'unattended cash (Transaction is cash)' (07 01)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '1' (ie Cardholder verification was not successful
- TVR byte 3, bit 7 = '1' (ie Unrecognized CVM) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder Verification was performed) received at 1st GENERATE AC.
- CVM Result = 3F 00 01

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2CJ.095.09 Second byte of CVR satisfied and CVM Code not understood by the terminal (10)

Test No: **2CJ.095.09** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal sets the 'Unrecognized CVM ' bit in the TVR to 1b, if the condition code is satisfied and CVM code is not understood by the terminal (RFU)

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Book 4 – Section 6.3.4 - Cardholder Verification Processing

Terminal Configuration (New format): NOT[Biometric] AND [Attended] AND [Cash]

Conditions:

- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM List is 'RFU' with the following conditions satisfied:
- condition is 'If manual cash (Transaction is cash)' (07 04)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '1' (ie Cardholder verification was not successful
- TVR byte 3, bit 7 = '1' (ie Unrecognized CVM) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder Verification was performed) received at 1st GENERATE AC.
- CVM Result = 3F 00 01

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2CJ.095.10 Second byte of CVR satisfied and CVM Code not understood by the terminal (11)

Test No: **2CJ.095.10** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal sets the 'Unrecognized CVM ' bit in the TVR to 1b, if the condition code is satisfied and CVM code is not understood by the terminal (RFU)

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Book 4 – Section 6.3.4 - Cardholder Verification Processing

Terminal Configuration (New format): NOT[Biometric] AND [Cashback]

Conditions:

- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM List is 'RFU' with the following conditions satisfied:
- condition is 'If purchase with cashback' (Transaction is purchase with cashback)' (07 05)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '1' (ie Cardholder verification was not successful
- TVR byte 3, bit 7 = '1' (ie Unrecognized CVM) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder Verification was performed) received at 1st GENERATE AC.
- CVM Result = 3F 00 01

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2CJ.095.11 Second byte of CVR satisfied and CVM Code not understood by the terminal (12)

Test No: **2CJ.095.11** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal sets the 'Unrecognized CVM ' bit in the TVR to 1b, if the condition code is satisfied and CVM code is not understood by the terminal (RFU)

Reference Book 3 - Section 10.5 - Cardholder Verification

EMV 4.4: Book 4 – Section 6.3.4 - Cardholder Verification Processing

Terminal Configuration (New format): NOT[Biometric] AND ([Goods] OR [Services] OR [Inquiry] OR [Transfer] OR [Payment] OR [Administrative] OR [CashDeposit])

Conditions:

- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM List is 'RFU' with the following conditions satisfied:
- condition is 'not unattended cash and not manual cash, and not purchase with cashback (Transaction is not cash and not purchase with cashback')' (07 02)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '1' (ie Cardholder verification was not successful
- TVR byte 3, bit 7 = '1' (ie Unrecognized CVM) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder Verification was performed) received at 1st GENERATE AC.
- CVM Result = 3F 00 01

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2CJ.095.12 Second byte of CVR not satisfied and CVM Code not understood by the terminal (13)

Test No: **2CJ.095.12** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure the terminal discards the CV Rule when the condition code is ‘if terminal supports’ and CVM code is not understood by the terminal (RFU).

Reference EMV 4.4: Book3 - Section 10.5 - Cardholder Verification
Book4 – Section 6.3.4 - Cardholder Verification Processing

Terminal Configuration (New format): [NA]

Conditions:

- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- Case 01: CVM List is 'RFU, if terminal supports (10 03) followed by fail CVM, always (00 00)’,
- Case 02: CVM List is 'RFU, if terminal supports (50 03) followed by fail CVM, always (00 00)’,

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '1' (ie Cardholder verification was not successful)
- TVR byte 3, bit 7 = '0' (ie Recognized CVM) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder Verification was performed) received at 1st GENERATE AC.
- CVM Result = 00 00 01 (fail CVM performed)

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2CJ.100.05 Non Supported CVR condition: If manual cash, and transaction is not manual cash

Revision: 44b

Objective: To ensure that terminal processes the next CVM in the list when the CVM condition is 'If manual cash' when transaction type is not manual cash.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Terminal ICS Options: [Unattended] AND ([Goods] OR [Services] OR [Cashback] OR [Inquiry] OR [Transfer] OR [Payment] OR [Administrative] OR [CashDeposit])

Conditions:

- Transaction type is not cash.
- LTsetting163:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '000000000000000000040000', 'Fail CVM If manual cash' (00 04) followed by 'Fail CVM, always' (00 00)

Procedure: Transaction is performed with TermSetting1 and LTsetting163

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '00 00 01'

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2CJ.112.00 Online PIN is the selected CVM and the online PIN processing is successful

Revision: 44b

Objective: To ensure that the terminal considers the CVM as successful, if enciphered PIN verified online is the selected CVM and Issuer answered that PIN presented is good.
To ensure that the PIN Pad enciphers the PIN upon entry for transmission in the authorization or financial transaction, if CVM is online PIN verification.
To ensure that the terminal sets the CVM Results byte 3 to 'unknown' when PIN is entered for online verification.

Reference EMV 4.4: Book 3 - Section 10.5.2 - Online PIN Processin
Book 4 - Section 6.3.4.5 - CVM Results

Terminal ICS Options: [Online Enciphered PIN]

Conditions:

- LTsetting167:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000200', 'Enciphered Online PIN verification, Always' (02 00)
- Issuer_response:
  - Issuer response indicates approval.

Procedure: Transaction is performed with TermSetting1 and LTsetting167

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 00 00'

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2CJ.158.00 Terminal Action Analysis & TVR and Issuer Action Code-Denial check requests an AAC

Test No: **2CJ.158.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal performs Action Analysis prior to the issuing of the first GENERATE AC command.
To ensure that the terminal issues a GENERATE AC requesting an AAC if for a bit that is set to 1b in the TVR, the corresponding bit in the IAC Denial is set to 1b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [NA]

Conditions:

- Terminal Action Codes have all bits set to 0b.
- Issuer Action Code Denial has one bit set to 1b and the corresponding TVR bit is set to 1b.
- Issuer Action Code Default has all bits set to 0b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by tester and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- The LT shall receive a first GENERATE AC command requesting an AAC.

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2CJ.159.00 Terminal Action Analysis & TVR and IAC-Denial check requests a TC (implied)

Test No: **2CJ.159.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal performs Action Analysis prior to the issuing of the first GENERATE AC command.
To ensure that the terminal issues a GENERATE AC requesting a TC if for a bit that is set to 1b in the TVR, the corresponding bit in the IAC Denial is set to 0b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline Only] OR [Offline/Online Capable]

Conditions:

- Terminal Action Codes have all bits set to 0b.
- Issuer Action Code Denial has one bit set to 0b and the corresponding TVR bit is set to 1b.
- Issuer Action Codes Online & Default have all bits set to 0b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester, and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- The LT shall receive a first GENERATE AC command requesting a TC

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2CJ.159.01 Terminal Action Analysis & TVR and IAC-Denial check requests an ARQC (implied)

Test No: **2CJ.159.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal performs Action Analysis prior to the issuing of the first GENERATE AC command.
To ensure that the terminal issues a GENERATE AC requesting an ARQC if for a bit that is set to 1b in the TVR, the corresponding bit in the IAC Denial is set to 0b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Online Only] OR [Offline/Online capable] OR ([Offline Only] AND [Offline Only Option 1])

Conditions:

- Terminal Action Codes Denial & Default have all bits set to 0b.
- Terminal Action Code Online has a bit set to 1b, in order that the Terminal request an ARQC
- Issuer Action Code Denial has one bit set to 0b and the corresponding TVR bit is set to 1b.
- Issuer Action Codes Online & Default have all bits set to 0b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester, and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- The LT shall receive a first GENERATE AC command requesting an ARQC

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2CJ.160.00 Terminal has online capability, TVR and Issuer Action Code-Online check requests an ARQC

Test No: **2CJ.160.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal issues a GENERATE AC requesting an ARQC, if the terminal has online capabilities and if for a bit set to 1b in the TVR, the corresponding bit in the IAC Online is set to 1b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Online Only] OR [Offline/Online capable] OR ([Offline Only] AND [Offline Only Option 1])

Conditions:

- Terminal Action Codes have all bits set to 0b.
- Issuer Action Codes Denial & Default have all bits set to 0b.
- Issuer Action Code Online has one bit set to 1b and the corresponding TVR bit is set to 1b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester, and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- The LT shall receive a first GENERATE AC command requesting an ARQC

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2CJ.160.01 Online only terminal requests an ARQC when not matching TAC-Online or IAC-Online

Test No: **2CJ.160.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal issues a GENERATE AC requesting an ARQC, if the terminal has online capabilities and if for a bit set to 1b in the TVR, the corresponding bit in the IAC Online is set to 0b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Online Only]

Conditions:

- Terminal Action Codes have all bits set to 0b.
- Issuer Action Codes have all bits set to 0b.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- The LT shall receive a first GENERATE AC command requesting an ARQC.
- TVR byte 2 - bit 3, is set to ‘0’, received at 2nd GENERATE AC.

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2CJ.161.00 Terminal has online capability, TVR and Online Action Codes check requests a TC (implied)

Test No: **2CJ.161.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal Issues a GENERATE AC requesting a TC, if terminal has online capabilities, and if for a bit set to 1b in the TVR, the corresponding bit in the IAC Online is set to 0b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline/Online capable] OR ([Offline Only] AND [Offline Only Option 1])

Conditions:

- Terminal Action Codes have all bits set to 0b.
- Issuer Action Codes Denial & Default have all bits set to 0b.
- Issuer Action Code Online has one bit set to 0b and the corresponding TVR bit is set to 1b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester, and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- The LT shall receive a first GENERATE AC command requesting a TC

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2CJ.162.00 Terminal is unable to go online, TVR and Issuer Action Code-Default check requests an AAC

Test No: **2CJ.162.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal issues a second GENERATE AC requesting an AAC, if online connection was requested after first GENERATE AC, terminal is unable to go online, and for a bit set to 1b in the TVR, the corresponding bit in the IAC-Default is set to 1b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline/Online Capable] OR ([Online Only] AND [Normal Default Action Code processing])

Conditions:

- Terminal is unable to go online
- Terminal Action Codes have all bits set to 0b.
- Issuer Action Code Denial has all bits set to 0b.
- LT returns ARQC to first GENERATE AC command.
- Issuer Action Code Default has one bit set to 1b and the corresponding TVR bit is set to 1b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester, and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- In all cases, the LT shall receive a second GENERATE AC command requesting an AAC

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2CJ.163.00 Terminal is unable to go online, TVR and Default Action Codes check requests a TC

Test No: **2CJ.163.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal issues a second GENERATE AC requesting a TC, if online connection was requested after first GENERATE AC, terminal is unable to go online, and for a bit set to 1b in the TVR, the corresponding bit in the IAC-Default is set to 0b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline/Online Capable] OR ([Online Only] AND [Normal Default Action Code processing])

Conditions:

- Terminal is unable to go online
- Terminal Action Codes have all bits set to 0b.
- Issuer Action Code Denial has all bits set to 0b.
- LT returns ARQC to first GENERATE AC command.
- Issuer Action Code Default has one bit set to 0b and the corresponding TVR bit is set to 1b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester, and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- The LT shall receive a second GENERATE AC command requesting a

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2CJ.164.00 Terminal has not online capability, TVR and Issuer Action Code-Default check requests an AAC

Test No: **2CJ.164.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal skips the check of Online Action Codes and issues a first GENERATE AC requesting an AAC, if terminal has not rejected the transaction, terminal has no online capabilities, and for a bit set to 1b in the TVR, the corresponding bit in the IAC-Default is set to 1b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline Only] AND [Offline Only Option 2]

Conditions:

- Terminal Action Codes have all bits set to 0b.
- Issuer Action Code Denial has all bits set to 0b.
- Issuer Action Code Default has one bit set to 1b and the corresponding TVR bit is set to 1b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester, and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- The LT shall receive a first GENERATE AC command requesting an AAC

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2CJ.165.00 Terminal has not online capability, TVR and Default Action Codes check requests a TC

Test No: **2CJ.165.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the terminal skips the check of Online Action Codes and issues a first or second GENERATE AC requesting a TC, if terminal has not rejected the transaction, terminal has no online capabilities, and for a bit set to 1b in the TVR, the corresponding bit in the IAC-Default is set to 0b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline Only] AND [Offline Only Option 2]

Conditions:

- Terminal Action Codes have all bits set to 0b.
- Issuer Action Code Denial has all bits set to 0b.
- Issuer Action Code Default has one bit set to 0b and the corresponding TVR bit is set to 1b.
- AIP of LT must be set to execute the function associated with the TAC bit selected by the tester, and the LT will be set so the executed function will fail
- LT returns TC at 1st GenAC

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- The LT shall receive a first GENERATE AC command requesting a TC

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2CJ.166.00 IAC-Denial is not present in the ICC

Test No: **2CJ.166.00** - *Revision* 43i - *Number of sub cases:* 0

Objective: To ensure that if the Issuer Action Code Denial does not exist in the card, the terminal uses a default.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline/Online capable] OR [Offline Only]

Conditions:

- Terminal Action Code Denial has all bits set to 0b.
- Terminal Action Code Online has all bits set to 0b.
- Terminal Action Code Default has all bits set to 0b.
- Issuer Action Code Denial does not exist in the card
- Issuer Action Code Online has all bits set to 0b.
- Issuer Action Code Default has all bits set to 0b.
- AIP of LT must be set to execute the function associated with the TAC bit selected by the tester and the LT will be set so the executed function will be fail, where the Default value of the TAC-Default defined in the ICS has the related bit of the function set to zero

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.

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2CJ.166.01 TAC-Denial is not present in the terminal

Test No: **2CJ.166.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Action Code Denial does not exist in the terminal, the terminal uses a default value with all bits set to 0b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): ([Offline/Online capable] OR [Offline Only]) AND [TAC Not Present]

Conditions:

- Terminal Action Code Denial is not present
- Terminal Action Code Online has all bits set to 0b.
- Terminal Action Code Default has all bits set to 0b.
- Issuer Action Code Denial has all bits set to 0b.
- Issuer Action Code Online has all bits set to 0b.
- Issuer Action Code Default has all bits set to 0b.
- AIP of LT must be set to execute the function associated with the TAC bit selected by the tester and the LT will be set so the executed function will be fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The LT shall receive a first GENERATE AC command requesting a TC

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2CJ.167.00 IAC-Online is not present in the ICC

Test No: **2CJ.167.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that If the Issuer Action Code Online does not exist in the card, the terminal uses a default value with all bits set to 1b.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline/Online capable] OR ([Offline Only] AND [Offline Only Option 1])

Conditions:

- Terminal Action Code Online has all bits set to 0b.
- Terminal Action Code Denial has all bits set to 0b.
- Terminal Action Code Default has all bits set to 0b.
- Issuer Action Code Online does not exist in the card
- Issuer Action Code Denial has all bits set to 0b.
- Issuer Action Code Default has all bits set to 0b.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester and the LT will be set so the executed function will fail

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC.
- The LT shall receive a first GENERATE AC command requesting an ARQC
- TVR byte 2 - bit 3 is set to ‘0’, received at 2nd GENERATE AC.

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2CJ.167.01 TAC-Online is not present in the terminal

Test No: **2CJ.167.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Action Code Online does not exist in the terminal, the terminal uses a default value.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): ([Offline/Online capable] OR ([Offline Only] AND [Offline Only Option 1])) AND [TAC Not Present]

Conditions:

- Terminal Action Code Online is not present
- Terminal Action Code Denial has all bits set to 0b.
- Terminal Action Code Default has all bits set to 0b.
- Issuer Action Code Denial has all bits set to 0b.
- Issuer Action Code Online has all bits set to 0b.
- Issuer Action Code Default has all bits set to 0b.
- AIP of LT must be set to execute the function associated with the TAC bit selected by the tester and the LT will be set so the executed function will be fail, where the Default value of the TAC-Default defined in the ICS has the related bit of the function set to zero.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- The LT shall receive a first GENERATE AC command requesting a cryptogram TC

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2CJ.168.00 IAC-Default is not present in the ICC and the Terminal is unable to go online

Test No: **2CJ.168.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the Issuer Action Code Default does not exist in the card, the terminal uses a default value with all bits set to 1b when the terminal is unable to go online.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline/Online Capable] OR ([Online Only] AND [Normal Default Action Code processing])

Conditions:

- Terminal Action Code Default has all bits set to 0b.
- Terminal Action Code Denial has all bits set to 0b.
- Terminal is unable to go online (no response received from issuer for instance)
- Issuer Action Code Default does not exist in the card
- Issuer Action Code Denial has all bits set to 0b.
- LT requests ARQC to first GENERATE AC command.
- AIP of LT must be set to execute the function associated with the IAC bit selected by the tester and the LT will be set so the executed function will fail for the second risk management

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The LT shall receive a second GENERATE AC command requesting an AAC.
- TVR byte 2 - bit 3 is set to ‘0’, received at 2nd GENERATE AC.

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2CJ.168.01 TAC-Default is not present in the Terminal and the Terminal is unable to go online

Test No: **2CJ.168.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the Terminal Action Code Default does not exist in the terminal, the terminal uses a default value.

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): ([Offline/Online Capable] OR ([Online Only] AND [Normal Default Action Code processing])) AND [TAC Not Present]

Conditions:

- Terminal Action Code Default is not present
- Terminal Action Code Denial has all bits set to 0b.
- Terminal Action Code Online has all bits set to 0b.
- Terminal is unable to go online (no response received from issuer for instance)
- Issuer Action Code Denial has all bits set to 0b.
- Issuer Action Code Online has all bits set to 0b.
- Issuer Action Code Default has all bits set to 0b.
- LT requests ARQC to first GENERATE AC command.
- AIP of LT must be set to execute a function which will fail for the second risk management (e.g. floor limit exceeded), where the Default value of the TAC-Default defined in the ICS has the related bit of the function set to zero.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion.
- The LT shall receive a second GENERATE AC command requesting a TC
- TVR byte 2 - bit 3 is set to ‘0’, received at 2nd GENERATE AC.

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2CJ.169.00 IAC-Default is not present in the ICC and the Terminal is offline only

Test No: **2CJ.169.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the Issuer Action Code Default does not exist in the card, the terminal uses a default value with all bits set to 1b when the terminal is offline only.

Reference EMV 4.4: Book 3 - Section 9.3 - Command Use
Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline Only] AND [Offline Only Option 2]

Conditions:

- Terminal Action Code Default has all bits set to 0b.
- Terminal Action Code Denial has all bits set to 0b.
- Terminal Action Code Online has all bits set to 0b.
- Issuer Action Code Default does not exist in the card
- Issuer Action Code Denial has all bits set to 0b.
- Issuer Action Code Online has all bits set to 0b.
- AIP must be set to execute a function associated with the IAC and this function will be failed by the LT (e.g. SDA is failed then the TVR byte 1 bit 7 is set to 1b)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The LT shall receive a first GENERATE AC command requesting an AAC

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2CJ.169.02 IAC-Default is not present in the ICC and the Terminal is offline only (2)

Test No: **2CJ.169.02** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that if the Issuer Action Code Default does not exist in the card, the terminal uses a default value with all bits set to 1b when the terminal is offline only

Reference EMV 4.4: Book 3 - Section 10.7 - Terminal Action Analysis

Terminal Configuration (New format): [Offline Only] AND [Offline Only Option 1]

Conditions:

- Terminal Action Code Default has all bits set to 0b.
- Terminal Action Code Denial has all bits set to 0b.
- Issuer Action Code Default does not exist in the card
- Issuer Action Code Denial has all bits set to 0b.
- Issuer Action Code Online has all bits set to 1b.
- AIP must be set to execute a function associated with the IAC and this function will be failed by the LT (e.g. SDA is failed then the TVR byte 1 bit 7 is set to 1).
- LT requests ARQC in First GENERATE AC

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The LT shall receive a second GENERATE AC command requesting an AAC
- TVR byte 2 - bit 3 is set to ‘0’, received at 2nd GENERATE AC.

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2CJ.178.00 Online Processing function execution

Revision: 44b

Objective: To ensure that the terminal performs the Online Processing function if the card returns an ARQC in the first GENERATE AC.

Reference EMV 4.4: Book 3 - Section 10.9 - Online Processing

Terminal ICS Options: [Online Only] OR [Offline/Online Capable]

Conditions:

- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’)

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- A Kernel Data Output shall be send after first Generate AC and before second Generate AC containing at least Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’).

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2CJ.203.01 Enhanced Combined DDA/AC failed when ICC responded with TC (2)

Revision: 44b

Objective: To ensure that the terminal declines the transaction if ICC has responded with TC and Enhanced Combined DDA/AC failed (2nd GENERATE AC)

Reference EMV 4.4: Book 2 - Section 6.6.2 - Dynamic Signature Verification, Book3 - Section 10.3 - Offline Data Authentication

Terminal ICS Options: [CDA] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- LTsetting1:
  - AIP indicates CDA is supported
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARCQC (‘80’)
- LT_response at second GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding a TC (‘40’)
  - Digital signature is not valid.
- Issuer_response:
  - The issuer does not respond to online processing request. Terminal unable to go Online.

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Data received at first GENERATE AC:
  - SDA not used (TVR_B1b7=‘0’).
  - DDA not used (TVR_B1b4=‘0’).
  - SDA not selected (TVR_B1b2=‘0’).
- Presence of Command of second GENERATE AC:
  - Command is received
- Terminal display: 'Declined' message
- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TSI_B1b8=‘1’).

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2CK.010.00 Card responds with an ARQC on first GENERATE AC

Revision: 44b

Objective: To ensure that if the card responds with an ARQC to first GENERATE AC and is permitted to do so and if terminal has online capability, the Kernel prepares a Kernel Data Output for an authorization or financial request message

Reference EMV 4.4: Book 3 - Section 9.3.1 - GENERATE AC (First Issuance)

Terminal ICS Options: [Online Only] OR [Offline/Online Capable]

Conditions:

- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC with Advice (‘08’)

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- A Kernel Data Output shall be send after first Generate AC and before second Generate AC containing at least Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’).

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2CM.020.00 Online PIN When PIN Try Limit is Exceeded and Terminal Supports GET DATA for PIN Try Counter

Revision: 44b

Objective: To ensure that the terminal allows a PIN to be entered for online verification even if the card's PIN Try Limit is exceeded

Reference EMV 4.4: Book 4 - Section 6.3.4.2 - Online CVM

Terminal ICS Options: [Get Data for PIN Try Counter] AND [Online Enciphered PIN] AND ([Offline Plaintext PIN] OR [Offline Enciphered PIN])

Conditions:

- LTsetting176:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '000000000000000004103440302 00', 'Plaintext PIN if terminal supports' (41 03) followed by 'Enciphered Offline PIN if terminal supports' (44 03) followed by 'Online Enciphered PIN always (02 00).
- LT_response at GET_DATA_PIN_TRY_COUNTER:
  - PIN Try Counter ‘9F17’ = 00

Procedure:

- Transaction is performed with TermSetting1 and LTsetting176

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - PIN Try Limit exceeded (TVR_B3b6='1')
- Terminal display: 'Enter PIN' Message

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2CM.020.01 Online PIN When PIN Try Limit is Exceeded and the Terminal does not support GET DATA for PIN Try Counter

Revision: 44b

Objective: To ensure that the terminal allows a PIN to be entered for online verification even if the card's PIN Try Limit is exceeded

Reference EMV 4.4: Book 4 - Section 6.3.4.2 - Online CVM

Terminal ICS Options: NOT[Get Data for PIN Try Counter] AND [Online Enciphered PIN] AND ([Offline Plaintext PIN] OR [Offline Enciphered PIN])

Conditions:

- LT_response at VERIFY:
  - LT returns status '63 C0' in response to VERIFY
- Case 01:
  - LTsetting177:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '00000000000000041030200', 'Plaintext PIN if terminal supports' (41 03), followed by 'Online Enciphered PIN always' (02 00).
- Case 02:
  - LTsetting178:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '00000000000000044030200', 'Enciphered Offline PIN if terminal supports' (44 03), followed by 'Online Enciphered PIN always' (02 00)
- Case 03:
  - LTsetting176:
    - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
    - CVM List has the value '00000000000000004410344030200', 'Plaintext PIN if terminal supports' (41 03) followed by 'Enciphered Offline PIN if terminal supports' (44 03) followed by 'Online Enciphered PIN always (02 00)

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting177
- Case 02: Transaction is performed with TermSetting1 and LTsetting178
- Case 03: Transaction is performed with TermSetting1 and LTsetting176

Pass Criteria:

- Parameter received at first or second GENERATE AC:

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2CM.020.01 Online PIN When PIN Try Limit is Exceeded and the Terminal does not support GET DATA for PIN Try Counter

- The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')

- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8='0')
  - Online PIN entered (TVR_B3b3='1')
  - PIN Try Limit exceeded (TVR_B3b6='1')
- Terminal display: 'Enter PIN' Message

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2CM.060.02 Financial Institution Terminal Supports TRM (3)

Revision: 44b

Objective: To ensure that the terminal supports Terminal Risk Management if it is a Financial institution Terminal

Reference EMV 4.4: Book 4 - Section 6.4 - Conditions for Support of Functions

Terminal ICS Options: [Terminal Type 1x] AND [Random Transaction Selection]

Conditions:

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Data received at first GENERATE AC:
  - Terminal Risk Management was performed (TSI_B1b4='1')

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2CM.077.00 Authorization Response Code when Referral is initiated by Issuer

Revision: 44b

Objective: To ensure that if the Authorization Response Code returned by the issuer, indicates a voice referral, a terminal does not modify the Authorization Response Code received

Reference EMV 4.4: Book 4 - Annex A6 - Authorization Response Code

Terminal ICS Options: ([Online Only] OR [Offline/Online Capable])

Conditions:

- LTsetting1:
  - CDOL2 request Authorization Response Code (8A)
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (80)
- Issuer_response:
  - Authorization Response Code returned in the response indicates a voice Referral.

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’) or an AAC (P1_b8='0' AND P1_b7=‘0’)
- Data received in second GENERATE AC:
  - Authorization Response Code shall be the same as the one received from Issuer

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2CM.081.00 Authorization Response Code when acceptance is forced

Revision: 44b

Objective: To ensure that if attendant forces the approval of a transaction, the terminal does not modify the Authorization Response Code

Reference EMV 4.4: Book 4 - Section 6.5.4 - Transaction Forced Acceptance

Terminal ICS Options: [Attended] AND [Forced Approval] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- LTsetting1:
  - CDOL2 request Authorization Response Code (8A)
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (80)
- Case 01:
  - Issuer_response:
    - Issuer response indicates decline.
- Case 02:
  - Issuer_response:
    - Issuer response indicates approval.
  - LT_response at second GENERATE AC
    - Cryptogram Information Data ‘9F27’ responding an AAC (‘00’)

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’) or an AAC (P1_b8='0' AND P1_b7=‘0’)
- Data received in second GENERATE AC:
  - Authorization Response Code shall be the same as the one received from Issuer

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2CM.086.00 Terminal Reading of Service Code on Magnetic Stripe

Revision: 44b

Objective: o ensure that the terminal displays a message 'Use Chip Reader' if the magstripe is read and the Service Code begins with '2' or '6', and if the terminal does not have a combined IC and magnetic stripe reader.

Reference EMV 4.4: Book 4 - Section 6.6 - Card Reading

Terminal ICS Options: [Magnetic Stripe Reader] AND NOT[Combined IC and Magnetic stripe reader]

Conditions:

- LTsetting1:
  - Service Code ‘5F 30’ not present in LT (ICC)
  - Magnetic is present and contain Service Code ‘5F 30’
- Case 01: Magstripe LT is read with Service Code beginning with '2'
- Case 02: Magstripe LT is read with Service Code beginning with '6'

Procedure:

- Magstripe shall be swiped first, and the ICC inserted following prompt.
- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Terminal display: 'USE CHIP READER' Message
- Data received in Kernel Data Output:
  - Shall not contain the Data Object: Service Code ’5F 30'

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2CM.087.00 Terminal Reading of Service Code on Magnetic Stripe

Revision: 44b

Objective: To ensure that the terminal switch to IC process if the magstripe is read and the Service Code begins with '2' or '6'

Reference EMV 4.4: Book 4 - Section 6.6 - Card Reading

Terminal ICS Options: [Combined IC and Magnetic stripe reader]

Conditions:

- LT is a combined magstripe and Chip ICC
- Case 01: Magstripe LT is read with Service Code beginning with '2'
- Case 02: Magstripe LT is read with Service Code beginning with '6'

Procedure:

- Transaction is performed with TermSetting1 and any LTsetting but Magstripe present

Pass Criteria:

- The terminal shall send EMV command to LT
- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')

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2CM.094.02 Current Date Management

Revision: 44b

Objective: To ensure that the terminal is able to accurately manage the current Transaction Date within the range 00-49 and range 50-99

Reference EMV 4.4: Book 4 - Section 6.7.3 - Date Management
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [Internal Date Management]

Conditions:

- LTsetting1:
  - CDOL1 requests Transaction Date and Transaction Time
- Case 01:
  - TermSetting1:
    - Terminal Date is set to 28/02/2039 23h59min
- Case 02:
  - TermSetting1:
    - Terminal Date set to 28/02/2051 23h59min

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Case 01
  - Data received at first GENERATE AC:
    - Transaction Date set to 01/03/2039
- Case 02
  - Data received at first GENERATE AC:
    - Transaction Date set to 01/03/2051

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2CM.094.03 Current Date Management

Revision: 44b

Objective: To ensure that the terminal is able to accurately manage the current Transaction Date within the range 00-49 and range 50-99

Reference EMV 4.4: Book 4 - Section 6.7.3 - Date Management
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [Online Only] AND NOT[Internal Date Management]

Conditions:

- LTsetting1:
  - CDOL1 requests Transaction Date and Transaction Time
- Case 01:
  - TermSetting1:
    - Current Date either in terminal host and in internal clock (if supported) is set to 28/02/2039 23h59min
- Case 02:
  - TermSetting1:
    - Current Date either in terminal host and in internal clock (if supported) is set to 28/02/2051 23h59min

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Case 01
  - Data received at first GENERATE AC:
    - Transaction Date set to 01/03/2039
- Case 02
  - Data received at first GENERATE AC:
    - Transaction Date set to 01/03/2051

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2CO.030.00 Terminal Issues a 2nd GENERATE AC Command (1)

Revision: 44b

Objective: To ensure that the terminal set Authorization Response Code to 'Unable to go online, offline accepted' and issues a second GENERATE AC indicating TC if the authorization response received is incorrect and if the terminal accepts the transaction.

Reference EMV 4.4: Book 4 - Section 12.2.3 - Authorisation Response Incidents

Terminal ICS Options: [Offline/Online Capable] OR ([Online Only] AND [Normal Default Action Code processing])

Conditions:

- LTsetting1:
  - CDOL2 request Authorization Response Code (8A)
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (80)
- Issuer_response:
  - no response received.

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at second GENERATE AC:
  - Authorization Response Code = 'Unable to go online, offline accepted (Y3)'

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2CO.031.00 Terminal Issues a 2nd GENERATE AC Command (2)

Revision: 44b

Objective: To ensure that the terminal set Authorization Response Code to 'Unable to go online, offline declined' and issues a second GENERATE AC indicating AAC if the authorization response received is incorrect if the terminal declines the transaction.

Reference EMV 4.4: Book 4 - Section 12.2.3 - Authorisation Response Incidents

Terminal ICS Options: [Online Only] OR [Offline/Online Capable]

Conditions:

- LTsetting179:
  - CDOL2 request Authorization Response Code '8A'
  - Issuer Action Codes Default '9F0D' = 00 40 00 00 00
  - Expiration Date '5F24' = 200101
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (80)
- Issuer_response:
  - no response received.

Procedure:

- Transaction is performed with TermSetting1 and LTsetting179

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting an AAC (P1_b8='0' AND P1_b7=‘0’)
- Data received at second GENERATE AC:
  - Authorization Response Code = 'Unable to go online, offline declined' (Z3)

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2CO.037.00 All data in the Messages are originated from chip

Revision: 44b

Objective: To ensure that terminal does not use magstripe data to populate an authorisation message or financial transaction message when a data is present in magstripe and not in ICC.

Reference EMV 4.4: Book 4 - Section 6.6 - Card Reading

Terminal ICS Options: [Combined IC and Magnetic stripe reader]

Conditions:

- LT is a combined magstripe and Chip ICC
- Track 2 Equivalent Data (tag '57') in LTsetting with it Service Code different from '2' and '6'
- Case 01: Magstripe LT is read with Service Code beginning with '2'
- Case 02: Magstripe LT is read with Service Code beginning with '6'

Procedure:

- Transaction is performed with TermSetting1 and any LTsetting but Magstripe present

Pass Criteria:

- The terminal shall send EMV command to LT
- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Data received in Kernel Data Output:
  - Track 2 Equivalent Data (tag '57') shall be the one retrieved from LT (with Service Code different from '2' and '6')

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2CS.014.00 Combined functions on Transaction Flow: Offline PIN and Advice with PIN Try Limit exceeded

Revision: 44b

Objective: To ensure that the Terminal is able to perform the transaction when advice 'PIN Try Limit exceeded' to the issuer is requested and PIN offline verification failed

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal ICS Options: [Offline Plaintext PIN]

Conditions:

- LTsetting136:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000100', 'Plaintext PIN verification Offline' (01 00).
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an AAC with Advice for 'PIN Try Limit exceeded' (0A)
- Action:
  - Cardholder enters a wrong PIN.

Procedure:

Transaction is performed with TermSetting1 and LTsetting136

Pass Criteria:

- Terminal display: 'Declined' message
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')

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2CS.014.01 Combined functions on Transaction Flow: Offline PIN and Advice with PIN Try Limit exceeded (2)

Revision: 44b

Objective: To ensure that the Terminal is able to perform the transaction when advice 'PIN Try Limit exceeded' to the issuer is requested and PIN offline verification failed

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal ICS Options: [Offline Plaintext PIN]

Conditions:

- LTsetting136:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000100', 'Plaintext PIN verification Offline' (01 00).
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an Advice for 'PIN Try Limit exceeded' (8A)
- Action:
  - Cardholder enters a valid PIN.

Procedure:

Transaction is performed with TermSetting1 and LTsetting136

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')

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2CS.014.02 Combined functions on Transaction Flow: Offline PIN and Advice with PIN Try Limit exceeded (3)

Revision: 44b

Objective: To ensure that the Terminal is able to perform the transaction when advice 'PIN Try Limit exceeded' to the issuer is requested and PIN offline verification failed

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal ICS Options: [Offline Enciphered PIN]

Conditions:

- LTsetting118:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000400', Enciphered PIN verification Offline' (04 00).
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an AAC with Advice for 'PIN Try Limit exceeded' (0A)
- Action:
  - Cardholder enters a wrong PIN.

Procedure:

Transaction is performed with TermSetting1 and LTsetting118

Pass Criteria:

- Terminal display: 'Declined' message
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - PIN entry required, PIN pad present, but PIN was not entered is not used (TVR_B3b4=‘0’)
  - PIN entry required and PIN pad present or working (TVR_B3b5=‘0’)
  - PIN Try Limit exceeded (TVR_B3b6=‘1’)

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2CS.014.03 Combined functions on Transaction Flow: Offline PIN and Advice with PIN Try Limit exceeded (4)

Revision: 44b

Objective: To ensure that the Terminal is able to perform the transaction when advice 'PIN Try Limit exceeded' to the issuer is requested and PIN offline verification was correct

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal ICS Options: [Offline Enciphered PIN]

Conditions:

- LTsetting118:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000400', Enciphered PIN verification Offline' (04 00).
- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an Advice for 'PIN Try Limit exceeded' (8A)
- Action:
  - Cardholder enters a valid PIN.

Procedure:

Transaction is performed with TermSetting1 and LTsetting118

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')

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2CS.015.00 Combined functions on Transaction Flow: SDA and Record length coded on 1 or 2 bytes

Test No: **2CS.015.00** - *Revision* 44b - *Number of sub cases:* 4

Objective: To ensure that the terminal supports Static Data Authentication and the records involved in the SDA calculation are coded on 1 or 2 bytes

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [SDA]

Conditions:

- Terminal contains the CA Public Key referenced in LT
- AIP of LT indicates Static Data Authentication is supported (AIP byte 1 bit 7 = ‘1’).
- Signed Static Application Data is good in LT
- Case 01: Record length listed in the AFL as participating in data authentication is coded on 1 byte (b8 = 0)
- Case 02: Record length listed in the AFL as participating in data authentication is coded on 2 bytes (81 xx)
- Case 03: The length of a Data Object participating in data authentication is coded on 1 byte (b8 = 0)
- Case 04: The length of a Data Object participating in data authentication is coded on 2 bytes (81 xx)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- Terminal Capabilities shall indicate Static Data Authentication support
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '1' (ie Offline Static Data Authentication selected) received at 1st GENERATE AC.

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2CS.015.01 Combined functions on Transaction Flow: SDA and Record length of proprietary file coded on 1 or 2 bytes

Test No: **2CS.015.01** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure that the terminal supports Static Data Authentication and the records of proprietary file involved in the SDA calculation are coded on 1 or 2 bytes.

Reference EMV 4.4: Book3 - Section 10.3 - Offline Data Authentication
Book3 – Annex B2 – Coding of the Length

Terminal Configuration (New format): [SDA]

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file, and listed in AFL and included in the data to be signed
- AIP of LT indicates Static Data Authentication is supported (AIP byte 1 bit 7 = ‘1’).
- EMV Data Object located in proprietary files will be TLV coded with record tag ‘70’
- Signed Static Application Data is good including in the computation the tag 70 and associated length of the record contained in the proprietary files
- Case 01: Record length of the proprietary file participating in data authentication is coded on 1 byte (b8 = 0)
- Case 02: Record length of the proprietary file participating in data authentication is coded on 2 bytes (81 xx)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT (in particular Static Data Authentication).

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Offline Static Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '1' (ie Offline Static Data Authentication selected) received at 1st GENERATE AC.

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2CS.015.02 Combined functions on Transaction Flow: DDA and Record length coded on 1 or 2 bytes

Test No: **2CS.015.02** - *Revision* 44b - *Number of sub cases:* 4

Objective: To ensure that the terminal supports Dynamic Data Authentication and the records involved in the DDA calculation are coded on 1 or 2 bytes

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [DDA]

Conditions:

- Terminal contains the CA Public Key referenced in LT
- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).
- Signed Dynamic Application Data is good in LT
- Case 01: Record length listed in the AFL as participating in data authentication is coded on 1 byte (b8 = 0)
- Case 02: Record length listed in the AFL as participating in data authentication is coded on 2 bytes (81 xx)
- Case 03: The length of a Data Object participating in data authentication is coded on 1 byte (b8 = 0)
- Case 04: The length of a Data Object participating in data authentication is coded on 2 bytes (81 xx)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- Terminal Capabilities shall indicate Dynamic Data Authentication supported
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC

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2CS.015.03 Combined functions on Transaction Flow: DDA and Record length of proprietary file coded on 1 or 2 bytes

Test No: **2CS.015.03** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure that the terminal supports Dynamic Data Authentication and the records of proprietary file involved in the DDA calculation are coded on 1 or 2 bytes.

Reference EMV 4.4: Book3 - Section 10.3 - Offline Data Authentication
Book3 – Annex B2 – Coding of the Length

Terminal Configuration (New format): [DDA]

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file, and listed in AFL and included in the data to be signed
- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).
- EMV Data Object located in proprietary files will be TLV coded with record tag ‘70’
- Signed Dynamic Application Data is good including in the computation the tag 70 and associated length of the record contained in the proprietary files
- Case 01: Record length of the proprietary file participating in data authentication is coded on 1 byte (b8 = 0)
- Case 02: Record length of the proprietary file participating in data authentication is coded on 2 bytes (81 xx)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC or an AAC at first or second GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Offline Dynamic Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.

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2CS.015.04 Combined functions on Transaction Flow: CDA and Record length coded on 1 or 2 bytes

Test No: **2CS.015.04** - *Revision* 44b - *Number of sub cases:* 4

Objective: To ensure that the terminal supports Combined DDA/AC Data Authentication and the records involved in the CDA calculation are coded on 1 or 2 bytes

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [CDA]

Conditions:

- Terminal contains the CA Public Key referenced in LT
- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 01: Record length listed in the AFL as participating in data authentication is coded on 1 byte (b8 = 0)
- Case 02: Record length listed in the AFL as participating in data authentication is coded on 2 bytes (81 xx)
- Case 03: The length of a Data Object participating in data authentication is coded on 1 byte (b8 = 0)
- Case 04: The length of a Data Object participating in data authentication is coded on 2 bytes (81 xx)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC
- Terminal Capabilities shall indicate Combined Data Authentication support
- TVR byte 1, bit 4 = '0' (ie Offline Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) contained in Kernel Data Output (this pass criteria apply only if CDA is requested)
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CS.015.05 Combined functions on Transaction Flow: CDA and Record length of proprietary file coded on 1 or 2 bytes

Test No: **2CS.015.05** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure that the terminal supports Combined DDA/AC Data Authentication and the records of proprietary file involved in the CDA calculation are coded on 1 or 2 bytes.

Reference EMV 4.4: Book3 - Section 10.3 - Offline Data Authentication
Book3 – Annex B2 – Coding of the Length

Terminal Configuration (New format): [CDA]

Conditions:

- An EMV Data Object is included in a record, located in a proprietary file, and listed in AFL and included in the data to be signed
- AIP of LT indicatesCombined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- EMV Data Object located in proprietary files will be TLV coded with record tag ‘70’
- Signed Dynamic Application Data is good including in the computation the tag 70 and associated length of the record contained in the proprietary files
- Case 01: Record length of the proprietary file participating in data authentication is coded on 1 byte (b8 = 0)
- Case 02: Record length of the proprietary file participating in data authentication is coded on 2 bytes (81 xx)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC or an AAC at first or second GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Offline Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Offline Combined Data Authentication succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) contained in Kernel Data Output (this pass criteria apply only if CDA is requested).
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CS.016.00 Combined Test: SDA Tag List and AFL in Static Data Authentication

Test No: **2CS.016.00** - *Revision* 44b - *Number of sub cases:* 8

Objective: To ensure that the terminal supports that SDA Tag List and record listed in AFL participate in the SDA

Reference EMV 4.4: Book 3 - Section 10.3 - Offline Data Authentication

Terminal Configuration (New format): [SDA]

Conditions:

- AIP of LT indicates Static Data Authentication is supported (AIP byte 1 bit 7 = ‘1’).
- SDA signature is good for each sub case
- Case 01: SDA Tag List contains tag '82' (AIP) and a record from an EMV file is listed in the AFL and participate in Static Data Authentication
- Case 02: SDA Tag List contains tag '82' (AIP) and a record from a proprietary file is listed in the AFL and participate in Static Data Authentication
- Case 03: SDA Tag List contains tag '82' (AIP) and a record from an EMV file and a record from a proprietary file is listed in the AFL and participate in Static Data Authentication
- Case 04: SDA Tag List contains tag '82' (AIP) and no records are listed as participating in Static Data Authentication
- Case 05: SDA Tag List is not present, and a record from an EMV file is listed in the AFL and participate in Static Data Authentication
- Case 06: SDA Tag List is not present and a record from a proprietary file is listed in the AFL and participate in Static Data Authentication
- Case 07: SDA Tag List is not present and a record from an EMV file and a record from a proprietary file is listed in the AFL and participate in Static Data Authentication

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Offline SDA succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.

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2CS.016.00 Combined Test: SDA Tag List and AFL in Static Data Authentication

- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '1' (ie Offline Static Data Authentication selected) received at 1st GENERATE AC.

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2CS.016.03 Combined Test: SDA Tag List and AFL in Dynamic Data Authentication

Test No: **2CS.016.03** - *Revision* 44b - *Number of sub cases:* 7

Objective: To ensure that the terminal supports that SDA Tag List and record listed in AFL participate in the DDA

Reference EMV 4.4: Book 3 - Section 10.3 - Offline Data Authentication

Terminal Configuration (New format): [DDA]

Conditions:

- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).
- DDA signature is good for each sub case
- Case 01: SDA Tag List contains AIP, and a record from an EMV file is listed in the AFL and participate in Dynamic Data Authentication
- Case 02: SDA Tag List contains AIP and a record from a proprietary file is listed in the AFL and participate in Dynamic Data Authentication
- Case 03: SDA Tag List contains AIP and a record from an EMV file and a record from a proprietary file is listed in the AFL and participate in Dynamic Data Authentication
- Case 04: SDA Tag List contains AIP and no records are listed as participating in Dynamic Data Authentication
- Case 05: SDA Tag List is not present, and a record from an EMV file is listed in the AFL and participate in Dynamic Data Authentication
- Case 06: SDA Tag List is not present and a record from a proprietary file is listed in the AFL and participate in Dynamic Data Authentication
- Case 07: SDA Tag List is not present and a record from an EMV file and a record from a proprietary file is listed in the AFL and participate in Dynamic Data Authentication

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT (In particular Dynamic Data Authentication).

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Offline DDA succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.

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2CS.016.03 Combined Test: SDA Tag List and AFL in Dynamic Data Authentication

- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 2 = '0' (ie Offline Static Data Authentication not selected) received at 1st GENERATE AC.

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2CT.008.00 Combined functions on Transaction Flow: SDA and PIN Encryption

Test No: **2CT.008.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Static Data Authentication and PIN Encryption during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [SDA] AND [Offline Enciphered PIN]

Conditions:

- AIP of LT indicates Static Data Authentication is supported (AIP byte 1 bit 7 = ‘1’).
- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM requires 'Enciphered PIN verification Offline' (04 00)

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '0' (ie Cardholder verification succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder verification was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Offline Static Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.

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2CT.009.00 Combined functions on Transaction Flow: DDA and PIN Encryption

Test No: **2CT.009.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Dynamic Data Authentication and PIN Encryption using ICC Public Key during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [DDA] AND [Offline Enciphered PIN]

Conditions:

- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).
- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM requires 'Enciphered PIN verification Offline' (04 00)
- All certificates and related calculations for PIN Encipherment are made from the ICC Public Key.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '0' (ie Cardholder verification succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder verification was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Offline Dynamic Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.

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2CT.009.01 Combined functions on Transaction Flow: DDA and PIN Encryption (2)

Test No: **2CT.009.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Dynamic Data Authentication and PIN Encryption using ICC PIN Encipherment Public Key during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [DDA] AND [Offline Enciphered PIN]

Conditions:

- AIP of LT indicates Dynamic Data Authentication is supported (AIP byte 1 bit 6 = ‘1’).
- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM requires 'Enciphered PIN verification Offline' (04 00)
- All certificates and related calculations for PIN Encipherment are made from the ICC PIN Encipherment Public Key.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 3, bit 8 = '0' (ie Cardholder verification succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder verification was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Offline Dynamic Data Authentication succeeded) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Combined Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) received at 1st GENERATE AC.

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2CT.010.00 Combined functions on Transaction Flow: CDA and PIN Encryption

Test No: **2CT.010.00** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Combined DDA/AC Data Authentication and PIN Encryption using ICC Public Key during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [CDA] AND [Offline Enciphered PIN] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM requires 'Enciphered PIN verification Offline' (04 00).
- All certificates and related calculations for PIN Encipherment are made from the ICC Public Key.
- IAC’s and TAC’s are set so that TC is requested at first and second Gen AC.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first GENERATE AC.
- TVR byte 3, bit 8 = '0' (ie Cardholder verification succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder verification was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Offline Combined Data Authentication succeeded or not used) contained in Kernel Data Output
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) contained in Kernel Data Output if CDA is requested
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.010.01 Combined functions on Transaction Flow: CDA and PIN Encryption

Test No: **2CT.010.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Combined DDA/AC Data Authentication and PIN Encryption using ICC PIN Encipherment Public Key during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [CDA] AND [Offline Enciphered PIN] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM requires 'Enciphered PIN verification Offline' (04 00)
- All certificates and related calculations for PIN Encipherment are made from the ICC PIN Encipherment Public Key.
- IAC’s and TAC’s are set so that TC is requested at first and second Gen AC.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first GENERATE AC.
- TVR byte 3, bit 8 = '0' (ie Cardholder verification succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder verification was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Offline Combined Data Authentication succeeded or not used) contained in Kernel Data Output
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) contained in Kernel Data Output if CDA is requested
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output

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2CT.010.02 Combined functions on Transaction Flow: CDA and PIN Encryption (3)

Test No: **2CT.010.02** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Combined DDA/AC Data Authentication and PIN Encryption using ICC Public Key during the same transaction.

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [CDA] AND [Offline Enciphered PIN] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM requires 'Enciphered PIN verification Offline' (04 00).
- All certificates and related calculations for PIN Encipherment are made from the ICC Public Key.
- IAC’s and TAC’s are set so that ARQC is requested at first Gen AC and a TC at second Gen AC.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TVR byte 3, bit 8 = '0' (ie Cardholder verification succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder verification was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Offline Combined Data Authentication succeeded or not used) contained in Kernel Data Output
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) contained in Kernel Data Output if CDA is requested
- TVR byte 2 - bit 3 is set to ‘0’, contained in Kernel Data Output.

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2CT.010.03 Combined functions on Transaction Flow: CDA and PIN Encryption (4)

Test No: **2CT.010.03** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Combined DDA/AC Data Authentication and PIN Encryption using ICC PIN Encipherment Public Key during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [CDA] AND [Offline Enciphered PIN] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
- CVM requires 'Enciphered PIN verification Offline' (04 00)
- All certificates and related calculations for PIN Encipherment are made from the ICC PIN Encipherment Public Key.
- IAC’s and TAC’s are set so that ARQC is requested at first Gen AC and a TC at second Gen AC.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TVR byte 3, bit 8 = '0' (ie Cardholder verification succeeded) received at 1st GENERATE AC.
- TSI byte 1, bit 7 = '1' (ie Cardholder verification was performed) received at 1st GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Offline Combined Data Authentication succeeded or not used) contained in Kernel Data Output
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1' (ie Offline Data Authentication was performed) contained in Kernel Data Output if CDA is requested
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.011.00 Combined functions on Transaction Flow: Issuer Authentication and Script Processing

Test No: **2CT.011.00** - *Revision* 44b - *Number of sub cases:* 3

Objective: To ensure that the Terminal is able to perform both Issuer Authentication and Issuer Script Processing during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [Online Only] OR [Offline/Online Capable]

Conditions:

- LT parameters are set so that transaction is performed on-line.
- AIP of LT indicates Issuer Authentication is supported (AIP byte 1 bit 3 = 1).
- Response message from the Issuer contains the following scripts:
- Case 01: one '71' script
- Case 02: one '72' script
- Case 03: one '71' script and one '72' script

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TVR byte 5 bit 6 = '0' (ie Script processing did not fail before final GENERATE AC (subcase 01 and 03)) received at 2nd GENERATE AC.
- TVR byte 5 bit 5 = '0' (ie Script processing did not fail after final GENERATE AC (subcase 02 and 03)) contained in Kernel Data Output
- TSI byte 1, bit 3 = '1' (ie Script processing was performed) contained in Kernel Data Output
- TVR byte 5 bit 7 = '0' (ie Issuer Authentication was successful) received at 2nd GENERATE AC.
- TSI byte 1, bit 5 = '1' (ie Issuer Authentication was performed) received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.011.01 Combined functions on Transaction Flow: Issuer Authentication and Script Processing (2)

Test No: **2CT.011.01** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Issuer Authentication and Issuer Script Processing with error during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [Online Only] OR [Offline/Online Capable]

Conditions:

- LT parameters are set so that transaction is performed on-line.
- AIP of LT indicates Issuer Authentication is supported (AIP byte 1 bit 3 = 1).
- Authorization response message contains one '71' Issuer script with the following commands.
- LT returns 90 00 to script command 1.
- LT returns 69 83 to script command 2.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TSI byte 1, bit 3 = '1' (ie Script processing was performed) received at 2nd GENERATE AC.
- TVR byte 5 bit 6 = '1' (ie Script processing failed before final GENERATE AC) received at 2nd GENERATE AC.
- TVR byte 5 bit 7 = '0' (ie Issuer Authentication was successful) received at 2nd GENERATE AC.
- TSI byte 1, bit 5 = '1' (ie Issuer Authentication was performed) received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’,received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.011.02 Combined functions on Transaction Flow: Issuer Authentication and Script Processing (3)

Test No: **2CT.011.02** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Issuer Authentication and Issuer Script Processing with error during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [Online Only] OR [Offline/Online Capable]

Conditions:

- LT parameters are set so that transaction is performed on-line.
- AIP of LT indicates Issuer Authentication is supported (AIP byte 1 bit 3 = 1).
- Authorization response message contains one '72' Issuer script with the following commands.
- LT returns 90 00 to script command 1.
- LT returns 69 83 to script command 2.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TSI byte 1, bit 3 = '1' (ie Script processing was performed) contained in Kernel Data Output
- TVR byte 5 bit 5 = '1' (ie Script processing failed after final GENERATE AC) contained in Kernel Data Output
- TVR byte 5 bit 7 = '0' (ie Issuer Authentication was successful) received at 2nd GENERATE AC.
- TSI byte 1, bit 5 = '1' (ie Issuer Authentication was performed) received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.011.03 Combined functions on Transaction Flow: Issuer Authentication and Script Processing (4)

Test No: **2CT.011.03** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Issuer Authentication and Issuer Script Processing with error during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [Online Only] OR [Offline/Online Capable]

Conditions:

- LT parameters are set so that transaction is performed on-line.
- AIP of LT indicates Issuer Authentication is supported (AIP byte 1 bit 3 = 1).
- Authorization response message contains one '71' Issuer script with the following commands.
- LT returns 90 00 to script command 1.
- LT returns 69 83 to script command 2.
- Authorization response message contains one '72' Issuer script with the following commands.
- LT returns 90 00 to script command 1.
- LT returns 69 83 to script command 2.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TSI byte 1, bit 3 = '1' (ie Script processing was performed) contained in Kernel Data Output
- TVR byte 5 bit 6 = '1' (ie Script processing failed before final GENERATE AC) received at 2nd GENERATE AC.
- TVR byte 5 bit 5 = '1' (ie Script processing failed after final GENERATE AC) contained in Kernel Data Output
- TVR byte 5 bit 7 = '0' (ie Issuer Authentication was successful) received at 2nd GENERATE AC.
- TSI byte 1, bit 5 = '1' (ie Issuer Authentication was performed) received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.011.04 Combined functions on Transaction Flow: Issuer Authentication and Script Processing (5)

Test No: **2CT.011.04** - *Revision* 44b - *Number of sub cases:* 0

Objective: To ensure that the Terminal is able to perform both Issuer Authentication and Issuer Script Processing with error during the same transaction

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [Online Only] OR [Offline/Online Capable]

Conditions:

- LT parameters are set so that transaction is performed on-line.
- AIP of LT indicates Issuer Authentication is supported (AIP byte 1 bit 3 = 1).
- Authorization response message contains one '71' Issuer script with the following commands.
- LT returns 90 00 to script command 1.
- LT returns 69 83 to script command 2.
- Authorization response message contains one '72' Issuer script with the following commands.
- LT returns 90 00 to script command 1.
- LT returns 90 00 to script command 2.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at second GENERATE AC.
- TSI byte 1, bit 3 = '1' (ie Script processing was performed) contained in Kernel Data Output
- TVR byte 5 bit 6 = '1' (ie Script processing failed before final GENERATE AC) received at 2nd GENERATE AC.
- TVR byte 5 bit 5 = '0' (ie Script processing succeeded after final GENERATE AC) contained in Kernel Data Output
- TVR byte 5 bit 7 = '0' (ie Issuer Authentication was successful) received at 2nd GENERATE AC.
- TSI byte 1, bit 5 = '1' (ie Issuer Authentication was performed) received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ received at 2nd GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.018.00 Combined Test: Get Processing and Generate AC with different response format, CDA

Test No: **2CT.018.00** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure that the terminal supports in the same transaction the response format 1 and 2 for Get Processing and GENERATE AC when performing Combined Data Authentication

Reference EMV 4.4: Book 3 - Section 6.5 - Commands

Terminal Configuration (New format): [CDA] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- IAC’s and TAC’s are set so that TC is requested at first and second Gen AC.
- LT returns TC a first GENERATE AC
- Case 01: Response to GET PROCESSING OPTIONS is in format 1 and Response to GENERATE AC is in format 2
- Case 02: Response to GET PROCESSING OPTIONS is in format 2 and Response to GENERATE AC is in format 2

Action: NA

Procedure: Application in LT is selected and transaction is performed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first GENERATE AC.
- TVR byte 1, bit 3 = '0' (ie Offline Combined Data Authentication succeeded or not used) contained in Kernel Data Output.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.018.01 Combined Test: Get Processing and Generate AC with different response format, CDA (2)

Test No: **2CT.018.01** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure that the terminal supports in the same transaction the response format 1 and 2 for Get Processing and GENERATE AC when performing Combined Data Authentication

Reference EMV 4.4: Book 3 - Section 6.5 – Commands

Terminal Configuration (New format): [CDA] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- LT parameters are set so that transaction is performed on-line
- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- IAC’s and TAC’s are set so that ARQC is requested at first Gen AC and TC at second Gen AC.
- Case 01: Response to GET PROCESSING OPTIONS is in format 1, Response to first GENERATE AC is in format 2 and Response to second GENERATE AC is in format 2
- Case 02: Response to GET PROCESSING OPTIONS is in format 2, Response to first GENERATE AC is in format 2 and Response to second GENERATE AC is in format 2

Action: NA

Procedure: Application in LT is selected and transaction is performed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC or AAC (if the terminal skip the IAC/TAC Default Action Codes) at second GENERATE AC
- TVR byte 1, bit 3 = '0' (ie Offline Combined Data Authentication succeeded or not used) contained in Kernel Data Output.
- TVR byte 1, bit 7 = '0' (ie Static Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (ie Dynamic Data Authentication not used) received at 1st GENERATE AC.
- TVR byte 2 - bit 3 is set to ‘0’ contained in Kernel Data Output.

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2CT.035.00 Combined functions on Transaction Flow: SDA, Keys remainder not present, Proprietary Data and EMV data

Test No: **2CT.035.00** - *Revision* 44b - *Number of sub cases:* 00

Objective: To ensure that the Terminal is able to perform correctly the transaction on the following LT profile: SDA, Keys remainder not present, Proprietary Data and EMV data

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [SDA]

Conditions:

- CA Public Key has a length of 1984 bits
- AIP of LT = '5C 00'
- Language Preference of LT = ' 6A 61 65 6E'
- Application Version Number of LT = '02 00'
- AFL of LT has a length of 20 bytes
- CDOL1 and CDOL2 of LT have a length of 128 bytes
- CVM List of LT = '00 00 00 00 00 00 00 00 41 03 1E 03 02 03 1F 03'
- A record of LT contain the proprietary tag '9F 54'
- A record of LT contain the proprietary tag 'DF 4F'
- A record of LT contain the proprietary tag '9F 55''
- Issuer Public Key has a length of 1744 bits
- Issuer Public Key Remainder is not present in LT

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 6 = '1' (ie ICC Data missing) received at 1st GENERATE AC.

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2CT.036.00 Combined functions on Transaction Flow: DDA, Keys remainder not present, Proprietary Data and EMV data

Test No: **2CT.036.00** - *Revision* 44b - *Number of sub cases:* 2

Objective: To ensure that the Terminal is able to perform correctly the transaction on the following LT profile: DDA, Keys remainder not present, Proprietary Data and EMV data

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [DDA]

Conditions:

- CA Public Key has a length of 1984 bits
- AIP of LT = '7C 00'
- Language Preference of LT = ' 6A 61 65 6E'
- Application Version Number of LT = '02 00'
- AFL of LT has a length of 20 bytes
- CDOL1 and CDOL2 of LT have a length of 128 bytes
- CVM List of LT = '00 00 00 00 00 00 00 00 41 03 1E 03 02 03 1F 03'
- A record of LT contain the proprietary tag '9F 54'
- A record of LT contain the proprietary tag 'DF 4F'
- A record of LT contain the proprietary tag '9F 55"
- Issuer Public Key has a length of 1744 bits
- ICC Public Key has a length of 1504 bits
- Case 01: Issuer Public Key Remainder is not present in LT
- Case 02: ICC Public Key Remainder is not present in LT

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first or second GENERATE AC.
- TVR byte 1, bit 6 = '1' (ie ICC Data missing) received at 1st GENERATE AC.

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2CT.037.00 Combined functions on Transaction Flow: CDA, Keys remainder not present, Proprietary Data and EMV data

Test No: **2CT.037.00** - *Revision* 44b - *Number of sub cases:* 02

Objective: To ensure that the Terminal is able to perform correctly the transaction on the following LT profile: CDA, Keys remainder not present, Proprietary Data and EMV data

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [CDA] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- CA Public Key has a length of 1984 bits
- AIP of LT = '7D 00'
- Language Preference of LT = ' 6A 61 65 6E'
- Application Version Number of LT = '02 00'
- AFL of LT has a length of 20 bytes
- CDOL1 and CDOL2 of LT have a length of 128 bytes
- CVM List of LT = '00 00 00 00 00 00 00 00 41 03 1E 03 02 03 1F 03'
- A record of LT contain the proprietary tag '9F 54'
- A record of LT contain the proprietary tag 'DF 4F'
- A record of LT contain the proprietary tag '9F 55"
- Issuer Public Key has a length of 1744 bits
- ICC Public Key has a length of 1504 bits
- IAC's and TAC's are set so that TC is requested at first Gen AC.
- If transaction is processed online, terminal is unable to go online.
- Case 01: Issuer Public Key Remainder is missing
- Case 02: ICC Public Key Remainder is missing

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC at first GENERATE AC.
- TVR byte 1, bit 6 = '1' (ie ICC Data missing) received either at 1st GENERATE AC or contained in Kernel Data Output.

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2CT.039.00 ICC supporting DDA Only

Test No: **2CT.039.00** - *Revision* 43i - *Number of sub cases:* 00

Objective: To ensure that the Terminal is able to perform correctly the transaction on the following LT profile

Reference EMV 4.4: Book 3 - Section 8 - Transaction Flow

Terminal Configuration (New format): [DDA]

Conditions:

- AIP of LT = '3C 00'
- AFL of LT indicates no records are signed (i.e. fourth byte of each entry = ‘00’)
- Signed Static Application Data (tag’93’) is not present in any record.

Action: NA

Procedure: Application in LT is selected and transaction is processed with LT.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC or an AAC at first or second GENERATE AC.
- TVR byte 1, bit 8 = '0' (i.e. Offline data authentication performed) received at 1st GENERATE AC.
- TVR byte 1, bit 6 = '0' (i.e. no ICC data missing) received at 1st GENERATE AC.
- TVR byte 1, bit 4 = '0' (i.e. DDA passed) received at 1st GENERATE AC.
- TSI byte 1, bit 8 = '1 (i.e. Offline data authentication was performed) received at 1st GENERATE AC.

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